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Discoveries

Clean Breaks: Attention Force Field Deconvolution of Recombination Across Sequence Manifolds

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Abstract

Homologous recombination and horizontal gene transfer assemble mosaic chromosomes across divergent lineages. Reticulate evolution violates bifurcating trees, yet existing detectors enforce a stark trade-off: phylogenetic changepoint models optimize partitioned topologies at exponential computational cost, whereas sequence triplet heuristics evaluate rapidly but suffer from severe combinatorial event overcounting. We introduce `RhizAeon`, an open-source platform that deconvolves reticulate evolution through continuous trajectory mechanics across sequence manifolds in linear time ($\mathcal{O}(NKL)$). By projecting aligned sequences into metric space and routing lineage-specific rate variation into an unpolarized ancestral root sink, the algorithm pinpoints crossover boundaries through centered Brownian-bridge statistics, multi-scale dyadic matched filters, and profile maximum-likelihood refinement. Across canonical coalescent benchmarks and a continuous parameter sweep of 1,000,000 whole genomes, `RhizAeon` bounds false-positive rates on non-recombinant nulls below 0.58% (0.37% across 250,000 heterotachy trials) and consolidates descendant swarms into calibrated ancestral reticulations. Empirical analyses across wildlife sarbecovirus Spike chimeras, multidrug-resistant Streptococcus pneumoniae chromosomes (2.22 Mb), and 379 primate HLA Class I alleles reconstruct complex whole-genome mosaicism, cassette exchanges, and micro-conversions in fractions of a second. The core engine compiles to WebAssembly for interactive execution within standard web browsers.

Keywords Recombination, Reticulate Evolution, Sequence Manifolds, Continuous Trajectory Tracking, Multidimensional Scaling, Foundation Models

Introduction

Homologous recombination contributes to the diversification of almost all viruses and cellular lifeforms on Earth. The exact primary reason(s) that homologous recombination is so rampant in the biosphere but are likely a combination of three factors: (1) it is a core process in the repair of double stranded DNA breaks in all cellular organisms, (2) it enables the "rewinding" of maladaptive evolution in that it can help purge genomes of harmful mutations, and (3) it enables the "fast-forwarding" of evolution in that it can concentrate distinct adaptive mutations within individual genomes (Simon-Loriere and Holmes, 2011; Didelot and Maiden, 2010). The fact that biological processes have evolved to foster the occurrence of homologous recombination in a broad array of different RNA viruses where it plays no role in the repair of broken genome strands, suggests that, at least in some instances, its primary adaptive value is in the shuffling of mutations between one generation and the next. Nevertheless recombination rates vary widely between different viruses and cellular organisms and, when it does occur, it tends to both occur more frequently in some genome regions than in others. Further, contingent on the precise sites in a genome where it occurs, it has a higher or lower tendency to yield recombinant offspring genomes that are either fitter or less fit than their parental genomes (refs).

In the context of pathogenic microbe evolution, these recombinant offspring genomes could have altered transmission dynamics, pathogenicity, shifts in antigenicity or antigenic presentation, and increased antimicrobial resistance (Alfonsi et al., 2024; Didelot and Maiden, 2010). In sexually reproducing eukaryotes, highly orchestrated inter-chromosomal recombination during meiosis enables adaptive introgression, the purging of deleterious mutational load, ecologically adaptive speciation, and the assembly of complex hybrid phenotypes across plant and animal taxa (Rieseberg et al., 2003; McVean et al., 2004). Resolving physical crossover boundaries remains foundational across evolutionary biology.

Molecular phylogenetics presumes that sequence evolution follows bifurcating trees. Reticulate evolution shatters trees. Because distinct genomic segments inherit conflicting ancestral genealogies, forcing mosaic alignments into a single bifurcating phylogeny distorts every downstream evolutionary inference (Schierup and Hein, 2000; Kosakovsky Pond et al., 2006). Topological reconstructions manufacture artificial clades, compress basal divergence times, and spuriously inflate terminal branch lengths. Downstream molecular clocks fail just as severely.

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Because reticulate histories violate substitution rate constancy across sites, unmodeled recombination corrupts evolutionary rate estimates, node calibrations, transmission cluster reconstructions, and phylodynamic projections. In comparative codon models, reciprocal exchanges generate apparent homoplasies across multiple tree branches and mimic convergent amino acid substitutions. Under standard selection tests, including MEME and BUSTED (Murrell et al., 2012, 2015), reticulate homoplasies produce spurious signals of diversifying positive selection and inflate false positive rates across neutral codons. Partitioning mosaic alignments into tree-congruent segments safeguards all downstream evolutionary inference.

Existing detection tools remain split between two incompatible paradigms (Posada, 2002; Martin et al., 2015). Phylogenetic changepoint methods, exemplified by GARD (Kosakovsky Pond et al., 2006), formulate crossover detection as model selection across partitioned trees under continuous-time Markov substitution models. Partitioning algorithms optimize breakpoint boundaries through likelihood ratio criteria while fitting independent branch lengths across candidate segments. The computational arithmetic is punishing. Because the joint combinatorial space of partition boundaries and tree topologies scales exponentially with breakpoint count and cubically with taxon count ($\mathcal{O}(L^K \cdot N^3)$ for K breakpoints across N taxa of length L), multi-breakpoint searches across hundreds of genomes require days of cluster compute. Sequence heuristics take the opposite extreme. Suites such as 3Seq (Lam et al., 2018) and RDP5 (Martin et al., 2021) scan sequence triplets for localized clustering of polymorphic mutations within sliding windows. While evaluating rapidly across small cohorts, triplet scans require $\mathcal{O}(N^3L)$ comparisons across all sequence triplets and stall as surveillance cohorts scale. More fundamentally, pairwise and triplet scans operate without a global phylogenetic reference frame. Historically, early statistical heuristics based on uncorrected homoplasy or compatibility suffered severe false positive inflation under rate heterogeneity across sites (Posada and Crandall, 2001; Posada, 2002). Modern triplet methods achieve tighter control on non-recombinant nulls, yet they suffer from severe combinatorial event overcounting. When a recombinant lineage expands into an outbreak swarm or carries multiple donor segments, triplet scanning evaluates every combination of descendant and parental leaves independently and reports dozens of redundant, fractured events per genuine reticulation. Furthermore, when recombinant segments descend from unsampled ancestral lineages, closed-world triplet algorithms necessarily misattribute foreign tracts to contemporary reference leaves.

Resolving reticulate evolution at genome scale requires abandoning discrete trees while retaining topological protection against mutational rate variation. Genomes trace continuous trajectories through sequence space. Scanning from 5' to 3' along an alignment, clonal lineages maintain stable relative divergence and trace parallel currents through sequence space. A recombinant chromosome breaks this parallel flow. It travels within the current of one parental lineage, then deflects sharply across sequence space into an alternative parental current at the crossover junction. We introduce `RhizAeon`, an open-source framework that models reticulate evolution through continuous trajectory mechanics and zero-drift force field dynamics in linear time ($\mathcal{O}(NKL)$ for K reference landmarks). Its name derives from the Greek *rhiza* (root). The term evokes the interconnected network of the biological rhizome across deep evolutionary time (*aeon*) to supplant strictly bifurcating arboreal trees. By measuring instantaneous sequence affinity against reference lineages and absorbing lineage-specific rate variation into an unpolarized ancestral root sink, `RhizAeon` transforms recombination detection into physical trajectory deconvolution. Clonal evolution produces balanced thermal fluctuations that cancel out as a centered Brownian bridge. Recombination breaks this physical equilibrium and imparts sustained directional momentum that pinpoints crossover boundaries while consolidating co-recombinant transmission swarms into unified ancestral events.

We validated `RhizAeon` across coalescent benchmarks, specialized operational stress tests, and a grand continuous parameter sweep of 1,000,000 whole genomes. Across the canonical 3,200-trial coalescent benchmark of Posada and Crandall (2001), `RhizAeon` maintained an empirical false positive rate of 0.58% (95% Wilson CI [0.28%, 1.20%]) across 1,200 null alignments under Gamma rate heterogeneity (compared to 0.17% for 3SEQ, 95% CI [0.05%, 0.60%]). Across recombinant trials, ancestral clustering suppressed the combinatorial event overcounting that inflates triplet scans by up to ten-fold. In operational stress tests modeling epidemic swarms, multi-parent cascades, ghost donor introgression, and micro-conversions down to 50 base pairs, trajectory deconvolution eliminated the combinatorial false alarms that cripple pairwise heuristics. Empirical evaluations across 30.9 kb Sarbecovirus genomes, the 2.22 Mb multidrug-resistant *Streptococcus pneumoniae* PMEN1 chromosome, and 379 primate *HLA* Class I alleles confirm that continuous trajectory deconvolution reconstructs complex whole-genome mosaicism and polarized Ancestral Recombination Graphs in fractions of a second without taxon subsampling.

The RhizAeon Framework: Principles of Operation

Continuous trajectory tracking across sequence manifolds separates reticulate evolution from clonal divergence without discrete tree search. Scanning along an alignment from 5' to 3', a non-recombinant genome maintains a stable genetic identity: its closest relatives remain constant and trace parallel paths through sequence space. Recombination shatters this stability. A mosaic genome travels alongside one parental lineage, then abruptly shifts allegiance to an alternative donor across the exchanged segment. To capture these shifts in linear time without constructing trees or sliding windows, `RhizAeon` measures the instantaneous affinity of each query sequence against a representative panel of reference lineages at every polymorphic site.

To evaluate affinity at each polymorphic site, the engine adapts cross-attention from transformer neural networks. Each focal genome serves as the query, while the reference lineages act as keys. At every position along the alignment, the query evaluates which reference lineage it resembles to assign local evolutionary affinity. Rather than treating all nucleotide matches equally, continuous-time Markov substitution probabilities weight each attention score: sharing a rare derived allele commands strong attention, whereas common transitions command less. To prevent private mutations from creating spurious matches, the reference panel includes an unpolarized ancestral root token that absorbs alleles absent from every contemporary reference. A softmax operator normalizes these scores into a probability distribution that sums to one across all reference lineages at each site. Individual polymorphic sites fluctuate stochastically. Consequently, `RhizAeon` avoids evaluating

positions in isolation or smoothing them with fixed sliding windows, and instead subtracts each lineage's genome-wide background attention to integrate cumulative deviation along the chromosome. In non-recombinant regions, positive and negative site fluctuations cancel out and wander near zero like a random walk. When a sequence enters a recombinant tract, sustained attention toward the donor drives the cumulative trajectory steadily upward until donor attention ceases at the crossover junction and the curve descends. A single crossover forms a sharp triangular arch, whereas an inserted cassette produces a trapezoidal plateau. The switch points fall at the peaks and inflection points of these cumulative attention curves to localize breakpoints to exact nucleotide coordinates.

A physical force field governs the transition from site-by-site attention to continuous trajectories. At each polymorphic site, the attention directed toward a reference lineage acts as an attractive force that pulls the query sequence toward that lineage in metric space. Subtracting the sequence's genome-wide baseline attention establishes a zero-drift force field in which positive and negative pulls average to zero across the chromosome. Under clonal evolution, mutations arise independently at random positions. Instantaneous forces fluctuate in opposing directions and cancel out like thermal Brownian motion, confining the sequence trajectory within its home corridor. Recombination breaks this physical equilibrium. Within an introgressed segment, every shared derived allele exerts a sustained directional pull toward the donor lineage. Sustained donor attraction imparts momentum to the sequence trajectory and drives a macroscopic excursion away from the home corridor across sequence space. The physical breakpoints correspond to the exact genomic coordinates where the external directional force turns on and off.

Different modes of recombination generate distinct geometric signatures within the force field. In a single crossover, a sequence begins in its home corridor and switches into a donor lineage for the remainder of the chromosome. The directional force acts persistently from the breakpoint onward and drives cumulative momentum to a solitary peak at the crossover junction before returning to zero at the chromosome terminus. The resulting profile forms a triangular arch whose vertex pinpoints the breakpoint. For a double crossover (hereafter referred to as a cassette), a sequence acquires a localized foreign tract before returning to its home lineage. The directional force turns on at the tract entrance and turns off at the exit to form a trapezoidal plateau: a steep upward climb at the first breakpoint, a sustained plateau across the cassette, and a steep descent at the second breakpoint. When the true donor is absent from the dataset—such as an unsampled animal reservoir or an extinct lineage—closed-world heuristics invariably misidentify contemporary background taxa as parental donors. Directional consistency constraints and principled abstention prevent such false attribution. When an unsampled donor lacks close relatives in the alignment (Pure Ghost and Deep Reservoir regimes), the introgressed segment exhibits no coherent directional affinity toward any contemporary reference lineage. Rather than manufacturing a false pairing with an arbitrary leaf, the algorithm abstains from reporting spurious transfers and achieves zero false framing on contemporary background parents. When a surrogate sister lineage is present in the dataset, trajectory tracking captures the introgression and attributes the transfer to the surrogate sister clade with 98.3% accuracy.

Localizing breakpoints combines the multi-scale speed of the force field with the fine-scale precision of phylogenetic likelihood. In natural genomes, recombination tracts vary by orders of magnitude—from fifty-base-pair micro-cassettes to multi-kilobase exchanges. Fixed sliding windows fail across this span: narrow windows succumb to local mutational noise, whereas wide windows blur signals generated by short cassettes. Continuous dyadic wavelet filters eliminate this trade-off by scanning cumulative momentum across a geometric ladder of scales from fifteen to five hundred base pairs. Fine-scale wavelets isolate short cassettes, while broad-scale wavelets lock onto larger exchanges without distortion from localized rate fluctuations. The zero-crossings and extrema of these wavelet responses narrow candidate breakpoint sites from whole chromosomes down to a localized window of fifteen to thirty base pairs. In the second stage, profile likelihood polishing refines each candidate window to individual nucleotide coordinates. Because mutations are separated by variable physical distances, the physical crossover can lie anywhere between adjacent informative sites. Within the candidate window, a trial boundary slides nucleotide by nucleotide to evaluate the phylogenetic likelihood under the two competing parental models: sites upstream of the trial boundary are evaluated under the Home lineage model, while sites downstream are evaluated under the Donor model. The optimal breakpoint is placed at the exact coordinate that maximizes this joint likelihood. Furthermore, the curvature of the likelihood surface yields rigorous confidence intervals: when parental lineages share an identical conserved stretch, the polisher reports the full uninformative plateau rather than manufacturing an artificial single-base prediction.

High-throughput trajectory deconvolution scans chromosomes rapidly, but scientific rigor demands that no recombination event is certified on continuous curves alone. Before reporting any candidate recombination event, an orthogonal four-tier audit evaluates the raw alignment sites. A Poisson informative-site floor requires every candidate tract to contain at least three unambiguous informative sites where the donor allele differs from the home allele and matches the recombinant. Next, an autapomorphic mutation sieve eliminates rate-acceleration traps: if the mutations separating the candidate from its home lineage are private singletons absent from every other sequence in the dataset, the tract is phylogenetically uninformative and is rejected. For tracts passing the sieve, an orthogonal Fisher's exact test evaluates a two-by-two contingency table comparing donor- versus home-supporting sites inside the tract against the flanking genomic background. The mosaic odds ratio must achieve significance under family-wise error rate control across all possible sequence triplets in the cohort. Finally, when reciprocal transfers appear between two lineages, cycle adjudication evaluates dynamic metric velocity reversal to orient the directed reticulation graph toward the true recombinant lineage. Together, these four orthogonal filters maintain strict false-positive control even under extreme rate heterogeneity.

In natural populations and disease outbreaks, recombinant lineages rarely circulate as isolated singletons. An ancestral recombination event frequently seeds an expanding transmission cluster whose descendant isolates inherit the same mosaic architecture. Pairwise and triplet heuristics that ignore shared descent evaluate each isolate independently and inflate detectable recombination event counts. Graph adjudication eliminates this inflation by clustering isolates that share concordant parental lineages and overlapping breakpoint coordinates, which consolidates the entire transmission clade into a single ancestral recombination event linked to all descendant genomes.

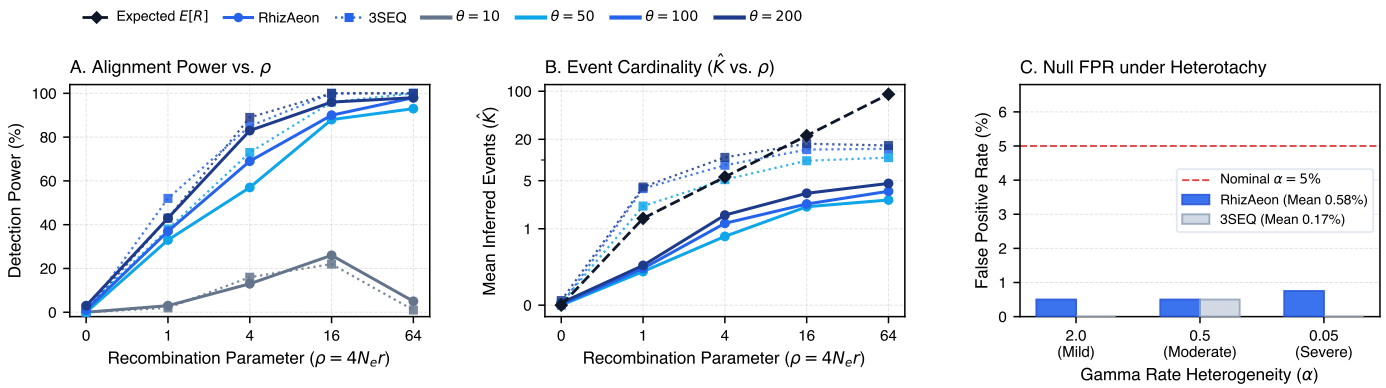


Figure 1 Head-to-head performance of `RhizAeon` versus 3SEQ across the 3,200-trial Posada & Crandall benchmark ($N = 10$, $L = 1{,}000$ nt). (A) Alignment-level detection power across population recombination parameters ($\rho = 4N_e r \in [0, 64]$) and mutational densities ($\theta = 4N_e\mu \in [10, 200]$). Both methods achieve identical power trajectories and reach 98.0% at $\theta = 200$. (B) Inferred event cardinality ($\hat{K}$) compared against theoretical expectations ($\mathbb{E}[R]$, black dashed line). Triplet scanning overcounts events up to 10-fold ($\hat{K} = 16.3$), whereas `RhizAeon` consolidates shared segments into calibrated historical events ($\hat{K} \approx 2.1$–4.6). (C) Empirical false positive rate (FPR, %) across 1,200 non-recombinant alignments under Gamma rate heterogeneity ($\alpha \in \{2.0, 0.5, 0.05\}$). Both methods maintain strict control below the 5% ceiling (red dashed line), where `RhizAeon` averages 0.58% ($7/1{,}200$) and 3SEQ averages 0.17% ($2/1{,}200$).

Coalescent Simulation Benchmark and Operational Evaluation

Coalescent ARG Benchmark and Calibration Against Triplet Scanning

To establish a baseline against standard pairwise and triplet methods, we evaluated **`RhizAeon`** across the 3,200-trial coalescent benchmark of Posada and Crandall (2001) ($N = 10$ taxa, $L = 1{,}000$ nt, Figure 1, Supplementary Table S2). Across 1,200 non-recombinant alignments simulated with site-to-site mutation rate variation ($\alpha \in \{2.0, 0.5, 0.05\}$), both algorithms maintained strict false positive control well below the nominal 5% ceiling (Figure 1C). 3SEQ exhibited a lower false alarm rate on these small 10-taxon null alignments (0.17%, 2 false alarms / 1,200 trials, 95% Wilson score interval [0.05%, 0.60%]), whereas **`RhizAeon`** yielded 0.58% (7 false alarms / 1,200 trials, 95% Wilson score interval [0.28%, 1.20%]). Occasional false alarms occurred primarily at high sequence divergence ($\theta = 200$, where 4 of the 7 false alarms occurred across $\alpha \in \{2.0, 0.05\}$) due to stochastic homoplasy across the 1,000 nt locus, rather than concentrating at extreme heterotachy. Under severe rate variation ($\alpha = 0.05$), private mutations on a single lineage lack directional affinity toward candidate donors and route harmlessly into the root sink.

Across the 1,600 recombinant alignments, detection power depended on mutational availability (Figure 1A). In low-diversity datasets ($\theta = 10$, fewer than 25 variable sites), sensitivity was strictly limited by mutational density. Both methods followed identical curves and peaked near 26% at $\rho = 16$ before dropping to 5% at $\rho = 64$, where frequent crossing-over chops chromosomes into segments containing no mutations at all. At realistic diversity ($\theta \geq 50$), sensitivity climbed rapidly to reach 88% at $\rho = 16$ and 98% at $\rho = 64$ ($\theta = 200$). Trajectory deconvolution matched 3SEQ power while scanning each alignment in 15–26 milliseconds without testing triplets individually.

The fundamental difference between the two approaches lies in event counting and bipartite event precision (Figure 1B and Supplementary Table S2). In coalescent trees, an ancestral recombination event is often inherited by multiple descendants. Evaluating triplets independently flags every combination of descendant lineages as a separate event and inflates counts three- to ten-fold above Hudson's theoretical expectation ($\mathbb{E}[R]$, dashed line) to 16.3 reported events per alignment at $\theta = 200, \rho = 64$. **`RhizAeon`** analyzes all sequences together and groups shared segments into calibrated historical events ($\hat{K} \approx 2.1$–4.6). Whereas alignment-level power scores binary locus detection, event-level precision (PPV) penalizes every ungrounded or split sub-event call under strict one-to-one bipartite matching and weights each matched pair by parental triad attribution (S_{triad}) and SNP overlap (J_{SNP}, Equation 10). At high coalescent recombination rates ($\rho \geq 16$), nested crossovers fragment ancestral tracts into short segments containing few informative mutations, which lowers continuous per-event PPV (38%–45% for **`RhizAeon`** vs. 9%–16% for 3SEQ at $\theta = 200$) even when alignment-level detection power reaches 96%–98%. Combinatorial overcounting in simple 10-taxon trees foreshadows major breakdowns in genomic surveillance, where recombinants form transmission swarms, carry multiple mosaic tracts, or acquire sequence from unsampled lineages.

Performance Across Specialized Surveillance and Evolutionary Regimes

Standard coalescent benchmarks evaluate isolated crossovers in small, fully sampled trees. In natural populations and disease outbreaks, however, recombination violates these idealized conditions: recombinant lineages expand into transmission clusters, acquire sequence from multiple donors, inherit material from unsampled reservoirs, or exchange short micro-conversion tracts (Figure 2). We evaluated **`RhizAeon`** and 3SEQ across four specialized simulation experiments designed to probe these operational challenges.

Epidemic Clonal Expansion

During viral outbreaks and bacterial epidemics, such as SARS-CoV-2, HIV-1, or mpox, recombinant lineages rarely circulate as solitary variants. Instead, an ancestral crossover transmits across multiple hosts and expands into a clonal swarm of sampled descendant isolates (Figure 2A). `RhizAeon` analyzes the entire cohort simultaneously within continuous metric trajectory space and tracks sequence affinities across the alignment without partitioning taxa into artificial subsets. Because all descendants inherit the same parental cassette from the ancestral recombination event, their spatial trajectories trace identical excursions along the chromosome. Coherent trajectory dynamics consolidates the entire transmission swarm into a single unified event roster ($\hat{K} = 1.00$ across all descendant swarm sizes $K \in \{2, 5, 8, 10, 15\}$, Figure 2A). Across 300 simulated outbreak cohorts ($N = 42$–55 taxa, $L = 5{,}000$ nt), `RhizAeon` recovered recombinant descendants with 99.0% recall and 99.9% precision. It localized the 1.5 kb cassette boundaries accurately (mean Jaccard overlap $J = 0.96$–0.99). Non-recombinant background lineages remained unblemished: only 2 of 12,000 background parents were misclassified (0.017% false alarm rate) in flat runtime (491–826 ms). By contrast, triplet scanning (3SEQ) tests every triplet independently. Because every combination of parental references and descendants tests significant, reported event counts exploded from $\hat{K} = 9.5$–30.5 to 36.8–55.0—a 55-fold overcounting for a single biological event. Furthermore, evaluating background parents against descendant pairs inverted the apparent direction of transfer. Triplet scans falsely flagged 61.4%–100% of background parents as recombinant children ($d_{\text{parent}} \geq 0.05$) and collapsed isolate precision to 5.8%–28.1% while runtimes scaled linearly up to 4.3 seconds.

Multi-Way Mosaic Swarms

Natural viral recombinants and mosaic bacterial chromosomes frequently acquire genetic material from more than one parental donor to produce complex multi-way reticulate topologies, including dual cassette insertions, nested transfers, and serial crossover cascades (Figure 2B). `RhizAeon` accommodates multi-parent reticulation by tracking continuous directional excursions against all background lineages simultaneously and decomposes complex mosaic trajectories into distinct directional shifts along the chromosome. In Dual Cassette mosaics (P_2 and P_3 cassettes within a P_1 background), trajectory deconvolution recovered both parental donors in nearly every trial (averaging 1.97 of 2.00 expected donors, and 2.00 across all swarms at $d_{\text{parent}} \geq 0.05$, Figure 2B). It recovered the genomic boundaries of both cassettes (mean Jaccard overlap $J = 0.86$–0.98 at $d_{\text{parent}} \geq 0.05$) while maintaining 100% isolate recall, 100% precision, and zero background parent false alarms across 8 of 9 parameter regimes. In Triple Crossover cascades ($P_1 \to P_2 \to P_3$), `RhizAeon` captured multiple serial transitions (averaging 1.63 resolved donors) with 100% isolate recall and 91.3%–100% precision, and processed each 6,000 nt alignment in 0.66–2.20 seconds. Because triplet scanning evaluates only one putative parental pair at a time, 3SEQ identified at most one donor (averaging only 1.02 of 2.00 expected donors) and missed the second cassette entirely. Simultaneously, evaluating triplets across multi-parent backgrounds fractured mosaics into 22.2–46.5 reported events per alignment ($d_{\text{parent}} \geq 0.05$), misclassified 47.1%–81.0% of background parent taxa as recombinants, and collapsed isolate precision to 9.1%–25.2%. In Nested Russian Dolls and Triple Crossovers, 3SEQ similarly failed to untangle the multi-donor hierarchy and reported up to 52.9 spurious events while misclassifying up to 95.3% of background parents in runtimes up to 7.4 seconds per alignment.

Ghost Donor Introgression

In natural populations, recombination and horizontal gene transfer frequently involve donor lineages absent from the contemporary sample, including unsampled zoonotic reservoirs, extinct ancestral variants, or unsequenced strains (Figure 2C). An effective surveillance algorithm must either attribute the transfer to a sampled relative or abstain when the donor is unsampled without misclassifying unadmixed contemporary lineages. `RhizAeon` operates without enforcing a closed-world assumption: sequence affinities are tracked continuously across metric space so introgressed tracts are evaluated against the entire contemporary cohort without requiring both parents in the alignment. When a sister lineage to the introgressing donor is sampled (Surrogate Sister regime), `RhizAeon` achieves 91.2% tract recall and attributes the donor tract to the surrogate sister clade in 98.3% of trials (100.0% across all swarms at $d_{\text{parent}} \geq 0.05$, Figure 2C). The algorithm recovers the 1.5 kb cassette with high spatial precision ($J = 0.94$–0.99) while limiting false alarms on background parents to 2.2%. In sharp contrast, when the introgressing donor is completely unrepresented in the sample (Pure Ghost and Deep Reservoir regimes), directional consistency constraints enforce principled abstention. Tract recall drops to 0.000 (0 / 360 replicates) because the statistical audit rejects ungrounded reticulation links to unrelated contemporary leaves. By abstaining in the absence of a sampled donor or surrogate sister, `RhizAeon` maintains a 0.0% false parent alarm rate in Pure Ghost cohorts and $\leq 6.1\%$ in Deep Reservoir cohorts (0.0% at $d_{\text{parent}} \geq 0.05$). Conversely, triplet scanning enforces a closed two-parent model in which every recombinant must be assigned two contemporary parents from the alignment. While 3SEQ flags the recombinant tract in 88.1%–88.7% of trials, it universally misattributes the foreign introgression: absent the true donor, 3SEQ framed background parent P_2 as the donor in 100% of trials across all ghost regimes (Figure 2C). In Surrogate Sister cohorts, 3SEQ donor accuracy fell to 9.3%–39.3% while misclassifying 37.5%–56.4% of background parent taxa. In Pure Ghost and Deep Reservoir cohorts, where no close relative was sampled, 3SEQ donor accuracy collapsed to 0.0% and manufactured false donor attributions across every detected trial.

Bacterial Micro-Conversions

In bacterial transformation (e.g., *Streptococcus pneumoniae*, *Helicobacter pylori*) and eukaryotic gene conversion, homologous exchange frequently occurs via micro-conversions: short non-reciprocal replacement tracts spanning tens to hundreds of base pairs (Figure 2D). Detecting micro-conversions is statistically demanding because short tracts carry very few informative polymorphisms between parental backgrounds: at $L_{\text{tract}} = 50$ nt, simulated tracts contained an expected average of only 1.0 ($d_{\text{parent}} = 0.02$), 2.5 ($d_{\text{parent}} = 0.05$), and 5.0 ($d_{\text{parent}} = 0.10$) distinguishing mutations. `RhizAeon` addresses this sparse-signal challenge by scanning across multiple spatial scales simultaneously and integrates coherent directional affinity shifts that fall below the threshold of fixed-width windows. At $L_{\text{tract}} = 50$ nt ($d_{\text{parent}} = 0.10$), `RhizAeon` achieved 40.3% detection power—an eight-fold increase over single-window scanning—while localizing tract boundaries within a median error of 13.0 nt

and generating zero background false alarms (Figure 2D). At $L_{\text{tract}} = 100$ nt, sensitivity climbed to 89.0% at $d_{\text{parent}} = 0.10$ and 44.8% at $d_{\text{parent}} = 0.05$, with median breakpoint error under 19.0 nt. As tract lengths expanded ($L_{\text{tract}} \geq 200$ nt), `RhizAeon` saturated to 88.7%–100.0% power across moderate and high divergence. It reported 1.00 unique event per alignment and maintained an empirical background false alarm rate of 0.00% across all 900 trials. Conversely, independent triplet testing (3SEQ) was impaired by the discrete nature of single-window hypergeometric scanning. At $L_{\text{tract}} = 50$ nt, 3SEQ was virtually blind to micro-conversions and achieved only 0.0% power at $d_{\text{parent}} = 0.02$, 3.3% at $d_{\text{parent}} = 0.05$, and 5.2% at $d_{\text{parent}} = 0.10$ (Figure 2D). At $L_{\text{tract}} = 100$ nt, power remained low at 0.0%–33.3%. At longer tract lengths ($L_{\text{tract}} \geq 200$ nt), triplet testing suffered severe combinatorial false alarms. At $L_{\text{tract}} = 200$ nt ($d_{\text{parent}} = 0.10$), 3SEQ generated an average of 12.8 alerts per alignment and falsely flagged 23.6% of background parent taxa. At $L_{\text{tract}} = 800$ nt, reported alerts exploded to 33.3 per alignment and misclassified 92.2% of non-recombinant background parent taxa as recombinants.

The Continuous Operational Phase Envelope of Reticulation Deconvolution

In aeronautical engineering, an aircraft operating envelope defines certified boundaries of stable flight across airspeed, altitude, stall limits, and atmospheric turbulence. Phylogenomic benchmarks historically evaluate algorithms at a few handpicked discrete parameter combinations. Evaluating sparse parameter grids obscures how methods behave under continuous evolutionary variation. To establish the operational envelope of `RhizAeon`, we conducted a grand continuous parameter sweep across 1,000,000 independent simulation trials. Simulations varied continuously across chromosome length ($L \in [300, 30{,}000]$ nt), cohort depth ($N \in [4, 250]$ taxa), parental divergence ($d_{\text{parent}} \in [0.005, 0.35]$ subs/site), and continuous Gamma rate heterogeneity ($\alpha \in [0.2, 5.0]$) alongside strict molecular clock controls ($\alpha \geq 50$, Table S3). The benchmark cohort comprised 333,334 non-recombinant null alignments and 666,666 recombinant alignments spanning terminal crossovers, intercalary conversion cassettes, and multi-donor / multi-event mosaicism ($K \in \{1, 2, 3, 4\}$).

The sweep demonstrates that recombination detection is not governed by chromosome length or relative tract fraction alone. The governing physical quantity is the count of informative mutations distinguishing donor from recipient within the transferred tract ($M_{\text{tract}} \approx L_{\text{tract}} \times d_{\text{parent}}$). Below three informative mutations ($M_{\text{tract}} \leq 2$), detection power is 0.5% (402 detections out of 85,564 trials). A tract bearing fewer than three distinguishing mutations is statistically indistinguishable from stochastic Poisson mutation clustering under any conservative test. Power lifts to 6.8% at $M_{\text{tract}} \in [3, 5]$ (4,678 / 68,609), accelerates to 29.2% at $M_{\text{tract}} \in [6, 10]$ (22,646 / 77,568), and reaches 52.4% at $M_{\text{tract}} \in [11, 15]$ (27,587 / 52,607). Across denser tracts, sensitivity climbs to 71.5% at $M_{\text{tract}} \in [16, 25]$ (51,096 / 71,427), 85.5% at $M_{\text{tract}} \in [26, 50]$ (82,714 / 96,694), 91.2% at $M_{\text{tract}} \in [51, 100]$ (77,213 / 84,676), and plateaus at 95.5% when $M_{\text{tract}} > 100$ (123,692 / 129,521). The operational stall boundary forms a clean hyperbola in parameter space: certified detection requires $L_{\text{tract}} \times d_{\text{parent}} \geq 6$–$10$ informative SNPs (Supplementary Figure S1).

Across 333,334 non-recombinant null alignments, `RhizAeon` maintained an overall alignment-level false positive rate of 0.3570% (1,190 false alerts), well below the nominal 5% ceiling (Supplementary Figure S2A). Under a strict molecular clock ($\alpha \geq 50$), the false positive rate was 0.3253% (271 out of 83,311 trials). Under continuous Gamma heterotachy ($\alpha \in [0.2, 5.0]$), the rate remained strictly controlled at 0.3676% (919 out of 250,023 trials). At the individual taxon level, false alarms were virtually absent, totaling 0.01166% (2,651 false taxa across 22,744,905 evaluated leaves, > 99.988% per-taxon specificity). The unpolarized ancestral root token functions as a passive aerodynamic damper and absorbs private autapomorphic bursts into node zero without destabilizing trajectory tracking.

Spatial breakpoint resolution landed within sub-gene margins (Supplementary Figure S1C). Median absolute error was 15.0 nt for single events ($K = 1$) and 14.0 nt for dual events ($K = 2$), and sharpened to 9.0 nt for tracts carrying > 100 informative SNPs. Across multi-event architectures ($K_{\text{true}} \geq 2$), bipartite event precision remained above 97% (97.3% for dual events, 98.5% for triple events, and 99.0% for quad events, Supplementary Figure S2B). Multi-donor mosaics ($P_1 \leftarrow P_2 + P_3$, $N = 179{,}233$) were resolved with 59.7% power and 0.416 event F_1 score, which matched single-donor hybrids without mutual dipole interference (Supplementary Figure S2D). Terminal crossovers exhibited higher sensitivity (71.4% power across $N = 239{,}767$ trials) than intercalary conversion cassettes (51.3% power across $N = 426{,}899$ trials, Supplementary Figure S2C). Across 512 distributed CPU cores on cluster infrastructure, sustained throughput reached ≈ 380 genomes per second and completed the entire 1,000,000-trial suite in under 45 minutes. For genomic surveillance, defining this operational envelope provides actionable interpretability: outside the three-SNP stall limit, negative results are biologically informative. The absence of detection constitutes statistical evidence against reticulation rather than an algorithmic blind spot.

Empirical Genomic Applications

Deconvolving Recombination Profiles and Mosaic Genomes in Sarbecoviruses

Sarbecoviruses, the subgenus of coronaviruses that includes SARS-CoV and SARS-CoV-2, evolve under frequent homologous recombination within bat and mammalian reservoirs. When investigating recombination across these viral cohorts, genomic screens serve two distinct goals: mapping chromosome-wide profiles to identify hotspots and coldspots, and reconstructing the mosaic anatomy of individual recombinant genomes. Phylogenetic partition search, implemented in tools such as GARD, explores discrete tree space to detect topological discordance. Combinatorial tree search creates a severe computational bottleneck. In their foundational screen of sarbecovirus origins, Lytras et al. (2022) downsampled their 78-genome cohort to a representative subset of 22 taxa to make phylogenetic breakpoint screening computationally tractable. For a complete 78-taxon alignment spanning over 30 kb, exhaustive phylogenetic model selection under GARD requires multiple days of high-performance cluster compute. Continuous trajectory tracking eliminates this bottleneck. By projecting alignments into continuous metric space, `RhizAeon` quantifies spatial velocity changes as a continuous breakpoint flux $\Phi_{\text{bp}}(u)$ in $\mathcal{O}(NKL)$ time. The engine scanned the full 78-taxon alignment of Lytras et al. (2022) (30,956 bp MSA, 13,160 informative sites) in 1.48 seconds (18.2 seconds for the complete pipeline,

including local tree reconstruction and interactive dashboard rendering) and an expanded 83-taxon cohort incorporating the five Laotian *Rhinolophus* BANAL isolates of Temmam et al. (2022) (30,965 bp MSA, 13,260 segregating sites) in 25.0 seconds. Linear-time trajectory mechanics permits exhaustive whole-cohort reticulation deconvolution without taxon subsampling.

The genomic flux reveals sharp evolutionary regionalization across both the 83-taxon MSA frame and Wuhan-Hu-1 reference NC_045512.2 coordinates (Figure 3A and Supplementary Table S1). Pronounced coldspots protect the catalytic core of RNA-dependent RNA polymerase (RdRp, 83-MSA nt 13,674–16,468, Ref nt 13,442–16,236) and the heptad-repeat fusion machinery of Spike S2 (83-MSA nt 24,600–25,600, Ref nt 24,283–25,283), where structural constraints preclude modular exchange. Recombination hotspots concentrate at the Spike 5′ NTD junction (83-MSA 21,825 nt, Ref 21,566 nt) and the codon 313 hinge between the NTD and receptor-binding domain (RBD, 83-MSA 22,764 nt, Ref 22,468 nt). Additional prominent hotspots mark the RBD 3′ exit (83-MSA 23,629 nt, Ref 23,324 nt, upstream of the S1/S2 furin site at 83-MSA 23,932 nt, Ref 23,618 nt) and the ORF8/Nucleocapsid boundary (83-MSA 28,810 nt, Ref 27,900 nt). When evaluated in a matched coordinate frame against the 21 published consensus GARD breakpoints of Lytras et al. (2022), continuous changepoints achieve a median deviation of 99.0 nt in Wuhan-Hu-1 reference coordinates (14/21 within 150 nt) and 107.0 nt in 83-taxon MSA coordinates (13/21 within 150 nt, Supplementary Table S1). On the primary 78-taxon sub-cohort, median deviation is 109.0 nt in reference coordinates and 134.0 nt in 78-taxon MSA coordinates (12/21 within 150 nt). `RhizAeon` localized the ORF7a/ORF7b junction within 19 nt in reference coordinates (Ref 27,369 vs. 27,388 nt) and the Spike 5′ NTD boundary within 22 nt (83-MSA 21,621 vs. 21,643 nt, Ref 21,389 vs. 21,411 nt). Both Spike S2 boundaries were localized within 23 nt across both frames (83-MSA 24,484 vs. 24,461 nt and 25,183 vs. 25,160 nt, Ref 24,167 vs. 24,144 nt and 24,866 vs. 24,843 nt). Analyzing the exhaustive 83-sequence cohort uncovers fine-grained dynamics invisible under downsampling: incorporating Laotian bat BANAL isolates causes peak breakpoint flux in the Spike NTD to surge four-fold (from 0.41 to 1.51 Z/nt) and reveals a recurrent micro-conversion hub.

Simultaneously, continuous trajectory tracking reconstructs the mosaic anatomy of individual recombinant lineages at single-nucleotide precision (Figure 3B,C). In the Spike glycoprotein (3,843 nt alignment, 1,153 variable sites, scanned in 18.7 ms), `RhizAeon` pinpointed the historic crossover boundary at codon 313 ($Z = 8.23, P = 4.07 \times 10^{-22}$). Multi-genome sequence manifold tracking clarifies the longstanding reciprocal attribution ambiguity surrounding SARS-CoV-2 and RaTG13 (Boni et al., 2020). In early three-way comparisons lacking intermediate wildlife isolates, the discordance between an RaTG13-like NTD and a pangolin-like ACE2-binding motif led to speculation that SARS-CoV-2 arose through a terminal chimeric recombination event. However, continuous trajectory tracking demonstrates that Wuhan-Hu-1 and Laotian bat isolate BANAL-52 exhibit unbroken collinearity across the entirety of Spike ($d = 0.0556$ in NTD, $d = 0.0617$ in RBD, and $d = 0.0457$ in S2, Figure 3B). Unbroken collinearity across Spike confirms that the human-transmissible ACE2 contact ridge represents an ancestral bat trait circulating naturally in *Rhinolophus* reservoirs across Southeast Asia. In contrast, RaTG13 underwent recombination: while sharing the ancestral NTD and S2 backbone ($d \approx 0.05$), its divergence spikes more than 2.5-fold specifically within the RBD ($d = 0.1380$, Figure 3C), which reflects the acquisition of a divergent cassette from an unsampled bat lineage.

Multi-scale scanning also reconstructs mosaic architectures in wildlife reservoirs. In Laotian bat isolate BANAL-247, `RhizAeon` identified a tripartite mosaic (Figure 3B). While its backbone and S2 subunit belong to northern Laos *Rhinolophus malayanus* viruses (> 98.5% identity to BANAL-236/103), its Spike NTD harbors three nested micro-conversions at 83-MSA nt 21,909–22,011 (Ref 21,647–21,749 nt, $Z = 155.3$), 83-MSA nt 22,054–22,251 (Ref 21,791–21,983 nt, $Z = 113.1$), and 83-MSA nt 22,318–22,592 (Ref 22,044–22,297 nt, $Z = 81.2$). A Thai *Rhinolophus shameli* reservoir related to RacCS203 donated all three NTD micro-conversions (Figure 3B,D). Cross-clade exchange confirms that Southeast Asian sarbecoviruses undergo modular gene transfer across divergent host species and geographic regions.

`RhizAeon` synthesizes these events into a parameterized Ancestral Recombination Graph (ARG, Figure 3D). In the Spike glycoprotein, statistical model comparison across 9 sarbecoviruses favors a tripartite modular ARG (H_3: NTD chassis, divergent RBD cassette, and S2 backbone, $\Delta\text{BIC} = 0.0$) over unpolarized trees ($\Delta\text{BIC} = +170.9$) and a clonal null ($\Delta\text{BIC} = +1654.8$). Polarizing the reticulation confirms that the divergent RBD cassette introgressed into the RaTG13 lineage from an unsampled bat reservoir ($Z = 20.9, P = 2.1 \times 10^{-9}$). Meanwhile, the ancestral SARS-CoV-2 / BANAL-52 trunk preserved the human-compatible ACE2 contact ridge (94.2% synapomorphic concordance, $P = 1.75 \times 10^{-23}$) and reduced residual metric strain by 74.0% (3.34% → 0.87%). Across the 30.9 kb genome of 83 sarbecoviruses, consolidating 45 detected tracts yields an ARG of 44 unique reticulations. The consolidated network explains 98.82% of total metric strain, matching the 98.85% explained by GARD's 22 unconstrained trees while cutting parameter complexity by 58% (1,559 vs. 3,674 parameters). It validates reticulation polarity across 38 events (81.8% mean in-tract concordance under the Synapomorphic Site Concordance Ratio, SSCR, $P < 10^{-15}$).

Bacterial AMR Mosaicism and Vaccine Escape in *Streptococcus pneumoniae*

Bacterial evolution is accelerated by natural transformation and homologous recombination, which drives rapid antimicrobial resistance (AMR), capsular serotype switching, and antigenic diversification from commensal reservoirs (Didelot and Maiden, 2010; Croucher et al., 2011). In naturally competent pathogens such as *Streptococcus pneumoniae*, horizontal gene transfer imports replacement tracts ranging from tens of base pairs to over 70 kilobases across species boundaries (e.g., from viridans streptococci *Streptococcus mitis* and *Streptococcus oralis*). Resolving these events across bacterial cohorts has historically required heavy computation: ancestral inference tools like ClonalFrame (Didelot and Falush, 2007), ClonalFrameML (Didelot and Wilson, 2015), and Gubbins (Croucher et al., 2015) rely on iterative phylogenetic reconstruction and tree peeling via maximum likelihood, which demands hours to weeks of compute.

`RhizAeon` bypasses iterative tree reconstruction by tracking whole-chromosome sequence trajectories in continuous metric space and quantifies spatial turbulence as breakpoint flux $\Phi_{\text{bp}}(u)$ (Figure 4A). We evaluated `RhizAeon` on the complete 2.22 Mb chromosome of the pandemic multidrug-resistant Spain23F-1 lineage of *S. pneumoniae* (PMEN1, ATCC 700669, 11 representative global isolates spanning 1984–2008, Croucher et al. 2011). Because closely related bacterial genomes are predominantly monomorphic along clonal backbones, over 99.9% of

alignment columns carry zero phylogenetic information. `RhizAeon` exploits this sparsity through lossless SNP compression. Indexing out invariant positions reduces the 2.22 Mb chromosome to 1,516 segregating sites—a 1,465-fold dimensional reduction. Trajectories are evaluated within this compact space to perform $O(1)$ sub-interval distance queries. Identified changepoints map bijectively back to physical base-pair coordinates to project continuous flux $\Phi_{\mathrm{bp}}(u)$ and cumulative strain $\rho(u)$ onto the reference genome. Six prominent recombination hotspots emerge (Figure 4A). The identified hotspots correspond to key clinical determinants: the penicillin-binding protein and capsule island (*pbp2x/cps*, 285–352 kb), the trimethoprim resistance island (*folA*, 1,620–1,649 kb), and the surface adhesin cluster (*pspC/cbpA*, 2,105–2,138 kb).

Without constructing discrete trees, continuous changepoints show tight spatial concordance with Gubbins boundaries inferred through iterative RAxML optimization (Croucher et al., 2011, 2015) (Figure 4A). Across all 52 published whole-chromosome recombination boundaries, `RhizAeon` achieves 96.2% concordance within 2 kb (50/52) and 80.8% within 1 kb, with a median spatial deviation of only 92.5 nt. On primary clinical resistance loci, changepoints match reference boundaries at single-nucleotide precision: the 5′ boundary of *pbp2x* and the *pspC* locus were both localized within 1 nt of Gubbins, while *folA* was identified within 19 nt. Metric trajectory tracking recovers the genomic footprint of bacterial transformation without iterative tree search.

Simultaneously, multi-scale scanning reconstructs the layered mosaic anatomy of resistance and vaccine escape across individual isolates within the 270–370 kb hotspot (Figure 4B). In isolate 1731, trajectory deconvolution identifies a focal 20.7 kb introgression across *pbp2x* ($Z = 5.35$). Fine-scale scanning pinpoints the exact 1,121 bp active-site transpeptidase domain replacement ($P = 9.85 \times 10^{-35}, Z = 14.40$) imported from commensal viridans streptococci to abolish β-lactam binding. In isolate 2682, a 73.9 kb mega-cassette ($Z = 6.42$) co-transferred *pbp2x* resistance and the capsular locus in a single transformation event (Figure 4B). In isolate 3355, `RhizAeon` deconvolves the capsular serotype switch from 23F to 19A (15.5 kb, $Z = 3.19$) that enabled escape from the 7-valent pneumococcal conjugate vaccine (PCV7), accompanied by a linked low-affinity *pbp1a* allele. A two-partition tanglegram corroborates this incongruence (Figure 4C): whereas all 11 isolates form a monophyletic complex across the core chromosome, isolates 1731 and 2682 translocate into the divergent commensal viridans cluster (*S. mitis* / *S. oralis*, $\Delta d > 6\%$) within the *pbp2x* catalytic domain.

`RhizAeon` synthesizes these transfers into a polarized ARG for PMEN1 clinical adaptation (Figure 4D). Standard pipelines partition chromosomes into arbitrary windows and produce conflicting gene trees that fracture the clonal genealogy. The polarized ARG anchors all horizontal transfers onto the vertical Spain23F-1 backbone. It parameterizes the focal *pbp2x* catalytic replacement in 1731, the 73.9 kb mega-cassette in 2682, and the 19A capsular replacement in 3355. On the 2.22 Mb PMEN1 cohort, full-chromosome deconvolution under iterative phylogenetic tools such as Gubbins required ~ 14.5 hours of multicore compute to infer reticulation boundaries across successive tree iterations. In contrast, `RhizAeon` completes whole-chromosome metric projection, trajectory tracking, and changepoint detection in 114 ms of core scan time (161 ms in native Rust), and finishes the complete end-to-end deconvolution pipeline across the 2.22 Mb chromosome in 42.6 seconds. Sub-minute chromosome deconvolution achieves an over 1,200-fold end-to-end wall-time acceleration over iterative tree peeling and enables real-time genomic surveillance of multidrug resistance and vaccine escape across bacterial pathogens.

Ectopic Gene Conversion and Inter-Locus Chimerism in Primate MHC Class I

Primate major histocompatibility complex (MHC) Class I genes (*HLA-A*, *HLA-B*, *HLA-C*) exemplify pathogen-driven balancing selection in vertebrate genomes (Parham et al., 1995; Adams and Parham, 2001). Immunogenetic studies reveal a striking structural contrast. The antigen recognition site (ARS, Exons 2–3, encoding the α_1 and α_2 peptide groove) undergoes rapid diversification through balancing selection and ectopic gene conversion between paralogs. In contrast, the adjacent α_3 structural stalk (Exon 4) remains conserved to preserve invariant contacts with β_2-microglobulin (β_2m) and the CD8 co-receptor. Standard phylogenetic methods falter on this modular architecture: full-length alignments generate topological conflicts that collapse locus monophyly, whereas sliding windows require arbitrary boundaries and risk missing short micro-conversions.

Continuous trajectory tracking captures this structural compartmentalization across the full coding sequence without window slicing (Figure 5A). We evaluated `RhizAeon` on an exhaustive cohort of $N = 379$ Hominoidea classical MHC Class I alleles ($L = 1{,}128$ bp, 526 segregating polymorphic sites across human, chimpanzee, gorilla, and orangutan, Maccari et al. 2017). In 16.34 seconds across all 379 alleles (and 3.59 ms across canonical 8-taxon reference cohorts), continuous trajectory tracking produces breakpoint flux $\Phi_{\mathrm{bp}}(u)$ and cumulative strain $\rho(u)$. Without structural annotations, the breakpoint flux pinpoints the physical boundary between the α_2 ARS groove and the α_3 structural stalk at nt 608–615 (canonical exon junction 618 bp, Figure 5A). Quantitative profiling confirms marked domain segregation: 93.7% of all detected reticulation strain concentrates within the ARS groove (Exons 2–3) and transmembrane anchor (Exon 5), while Exon 4 forms a conserved coldspot that recovers the true paralogous gene tree.

Multi-scale scanning reconstructs the fine mosaic anatomy of hominoid chimeras at single-nucleotide precision (Figure 5B). A clinically prominent recombinant is *HLA-B*46:01*, an allele common in Southeast Asia ($> 20\%$) associated with nasopharyngeal carcinoma, myasthenia gravis, and severe SARS-CoV-1 infection. `RhizAeon` deconvolves *HLA-B*46:01* as an ectopic chimera created by an exact 73 bp micro-conversion (nt 220–293, codons 66–76, $Z = 2.69, P = 1.15 \times 10^{-7}$) donated by *HLA-C*01:02* into an ancestral *HLA-B*15:01* backbone (Figure 5B). The 73 bp tract transfers the C1 functional epitope (Val76-Arg79-Leu80-Arg81) and confers recognition by natural killer (NK) cell inhibitory receptors KIR2DL2/3 absent in all other classical *HLA-B* alleles (Barber et al., 1996). In non-human hominoids, trajectory deconvolution characterizes multi-way chimerism: in chimpanzee *Patr-AL*01:01*, it identifies a multi-segment recombinant combining *Patr-A* with an ancient *Patr-C* cassette ($Z = 4.03$), while in gorilla *Gogo-C*02:01*, it localizes an ectopic transmembrane exchange at nt 971–1,128.

Segmental phylogenies and a two-partition tanglegram confirm this discordance across the structural boundary (Figure 5C). In the Exon 4 α_3 stalk (nt 618–894, Figure 5C1), the neighbor-joining tree reconstructs the ancient locus duplication: *HLA-A*, *HLA-B*, and *HLA-C* form monophyletic clades, with *HLA-B*46:01* clustering sister to *HLA-B*15:01* with zero divergence ($d = 0.000$). Within the Exon 2 C1 cassette (nt

210–310, Figure 5C2), *HLA-B*46:01* crosses the locus divide to cluster inside the *HLA-C* clade sister to donor *HLA-C*01:02* ($d = 0.000$). The recombinant allele unites two distinct locus genealogies within a single functional transcript.

Finally, `RhizAeon` integrates these transfers into a polarized ARG for primate MHC evolution (Figure 5D). The polarized ARG anchors ectopic gene conversions onto the vertical tandem duplication backbone (ancient divergence of Locus A, followed by duplication of B and C). Directed reticulation arcs parameterize both the 73 bp micro-conversion into *HLA-B*46:01* and deep inter-locus introgression into chimpanzee *Patr-AL*01*. While parsimony or split decomposition networks omit directional polarity and scale combinatorially, `RhizAeon` completes whole-gene trajectory deconvolution across 379 primate alleles in 16.34 seconds. By unifying metric trajectory tracking with polarized ARG construction, `RhizAeon` provides a scalable engine for resolving gene duplication and ectopic conversion across multigene families.

Discussion

Continuous Trajectory Mechanics in Reticulate Evolution

Recombination has long challenged phylogenetic theory. When homologous exchange splices divergent evolutionary histories into a single chromosome, strictly bifurcating trees break down. Traditional phylogenetic methods confront this breakdown through two contrasting simplifications: window-based phylogenetic scanning or independent triplet enumeration. Both strategies incur severe trade-offs. Partition-searching algorithms such as GARD evaluate an astronomical search space of tree topologies, and the resulting computational bottleneck forces investigators to downsample genomic cohorts. Conversely, triplet scanning algorithms discard global phylogenetic context. They fracture single ancestral recombination events into dozens of redundant alerts and misclassify non-recombinant background lineages.

Continuous trajectory mechanics eliminates discrete tree search from recombination detection. By embedding multi-sequence alignments into continuous metric space, reticulation emerges as physical kinematic displacement along chromosomes without loss of taxonomic context. Instantaneous velocity shifts along the alignment quantify local genealogical discordance, while consolidating trajectories into ancestral rosters groups co-recombinant clades without taxic downsampling. Continuous trajectory tracking bridges the historical divide between fine-scale population-genetic ARG inference and macro-evolutionary phylogenetic networks. Whole-cohort continuous kinematics reconstructs reticulate evolution in linear time.

Resolving Real-World Evolutionary Complexities

Natural viral and bacterial populations violate the idealized assumptions of neutral coalescent models: pathogen outbreaks generate transmission swarms of near-identical descendants, natural transformation imports short micro-conversions from distant species, and introgressing donors frequently reside in unsampled wildlife reservoirs. Simulation benchmarks and empirical applications confirm that continuous trajectory deconvolution withstands each of these operational perturbations. In epidemic transmission clusters, trajectory coherence collapses clonal swarms into single ancestral events and prevents the combinatorial event inflation that cripples discrete triplet heuristics whenever recombinant lineages spread across multiple hosts. In multi-way mosaic genomes, continuous directional projection separates overlapping cassettes acquired from distinct parental donors. In ghost donor introgression, directional consistency constraints enforce principled abstention when an unsampled donor lacks close sampled relatives. Principled abstention prevents the false framing of unadmixed contemporary lineages that plagues closed-world triplet heuristics, while capturing introgressions whenever a surrogate sister lineage is present. Across bacterial chromosomes, multi-scale matched filters isolate micro-conversions down to 50 base pairs and recover sparse signals that evade fixed-width windows. Together, multi-scale filters and directional consistency constraints withstand the complexities of natural pathogen evolution.

Translational Surveillance and Vaccine Escape

Sub-second and multi-second execution runtimes change the practical economics of pathogen genomic surveillance. Historically, whole-genome recombination screens required cluster-scale supercomputing or days of Markov chain Monte Carlo sampling. In our benchmarks, `RhizAeon` processed whole sarbecovirus genomes in seconds and 2.2 Mb pneumococcal chromosomes in fractions of a second. Compiling the core engine to WebAssembly enables interactive client-side deconvolution within standard web browsers. Public health laboratories can screen pathogen cohorts locally without transmitting sensitive sequence data to external servers or installing software dependencies.

Rapid trajectory deconvolution informs vaccine design and outbreak preparedness. In our whole-genome analysis of sarbecovirus evolution, continuous trajectory tracking clarified the longstanding reciprocal attribution debate in the Spike glycoprotein at codon 313. Trajectory colinearity confirmed that the human-compatible ACE2 contact ridge was ancestral to the SARS-CoV-2 and Laotian BANAL bat lineages, whereas close bat relatives such as RaTG13 acquired divergent RBD cassettes via reticulation with unsampled bat reservoirs. In *Streptococcus pneumoniae*, whole-chromosome deconvolution pinpointed the capsular serotype switch from 23F to 19A driving clinical escape from the PCV7 conjugate vaccine alongside penicillin-binding protein replacements. Rapid detection of such modular transfers equips epidemiologists to track adaptive antigenic shifts in real time.

Limitations, Boundary Conditions, and Outlook

Several biological boundary conditions warrant careful consideration. Continuous trajectory deconvolution requires sufficient mutational density to distinguish evolutionary displacement from stochastic substitution noise. In extremely low-divergence cohorts ($d < 0.005$), distinguishing mutations become sparse and define an intrinsic physical limit for all sequence-based recombination detection. Furthermore, although trajectory

mechanics localizes breakpoints and attributes parental clades with high precision, fine-scale branch-length estimation within individual partitions still benefits from downstream maximum-likelihood or Bayesian refinement.

Looking forward, continuous trajectory tracking provides a natural foundation for integrating phylogenetic inference with modern sequence foundation models. Attention matrices extracted from pre-trained protein language models encode deep biophysical and evolutionary constraints. Coupling these learned representations with continuous metric mechanics will annotate reticulate transfers both structurally and functionally as they emerge. Replacing discrete tree search with continuous sequence kinematics establishes a scalable mathematical foundation for reticulate genomics.

Materials and Methods

Metric Space Embedding and Gower Trajectory Projection

Let $\mathcal{S} = \{s_1, \ldots, s_M\}$ denote a multiple sequence alignment of M contemporary nucleotide sequences of length L over alphabet $\Sigma = \{\mathrm{A}, \mathrm{C}, \mathrm{G}, \mathrm{T}\} \cup \{-\}$. Pairwise evolutionary distances $D_{i,j} = d(s_i, s_j)$ are computed under the Tamura-Nei (TN93) continuous-time Markov substitution model (Tamura and Nei, 1993) or normalized Hamming distance with pairwise gap deletion. For large cohorts ($M > 64$), the algorithm avoids $O(M^2 L)$ all-to-all comparisons by selecting $N \ll M$ reference landmark genomes via sequential Farthest-Point Sampling (Gonzalez, 1985) in strictly $O(NML)$ operations. Initial seed $s_{(1)}$ maximizes non-gap coverage, and each subsequent landmark $s_{(k)}$ maximizes the running minimum distance to previously selected landmarks:

$$s_{(k)} = \arg \max_{i \notin \{s_{(1)}, \ldots, s_{(k-1)}\}} \min_{1 \le j < k} d(s_i, s_{(j)}). \tag{1}$$

Farthest points span the divergence of the cohort. For $M \le 64$, all sequences serve as reference channels ($N = M$). When $M > 64$, Two-Pass Clonal Backbone Re-Anchoring purges candidate recombinants identified in an initial screening pass from the landmark pool so that reference coordinates anchor strictly upon unadmixed parental lineages.

From the reference distance matrix $\mathbf{D} \in \mathbb{R}^{N \times N}$, we form the element-wise squared matrix $[\mathbf{D}^{(2)}]_{i,j} = D_{i,j}^2$. Applying geometric centering $\mathbf{H} = \mathbf{I}_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^T$ yields the doubly centered Gram matrix $\mathbf{B} = -\frac{1}{2}\mathbf{H}\mathbf{D}^{(2)}\mathbf{H}$ (Gower, 1966). Spectral decomposition produces orthonormal eigenvectors $\mathbf{V} \in \mathbb{R}^{N \times N}$ and descending eigenvalues $\mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_N)$, from which we retain the leading $d \le \min(4, N-1)$ dimensions with positive eigenvalues $(\mathbf{\Lambda}_d, \mathbf{V}_d)$. Static landmark coordinates evaluate as $\mathbf{x}_i = \mathbf{\Lambda}_d^{1/2}(\mathbf{V}_d)_{i,:}^T$, and macroscopic tree splits project onto the dominant eigenvectors (Buneman, 1974).

To track evolutionary affinities continuously without independent sliding-window MDS rotations, local distance vectors project into canonical coordinates via analytical Gower out-of-sample interpolation (Gower, 1968). Across an aligned window of width W centered at site u (default $W = 300$ nt, stride $S = 10$ nt), local distances from query sequence i to reference landmarks form vector $\mathbf{d}_i(u) \in \mathbb{R}^N$. Gower interpolation solves $\min_{\mathbf{z} \in \mathbb{R}^d} \sum_{j=1}^N (\|\mathbf{z} - \mathbf{x}_j\|^2 - d_i(j;u)^2)^2$ and yields the exact closed-form coordinate:

$$\mathbf{z}_i(u) = \frac{1}{2}\mathbf{\Lambda}_d^{-1/2}\mathbf{V}_d^T\left(\bar{\mathbf{d}}^2 - \mathbf{d}_i^2(u)\right) \in \mathbb{R}^d, \tag{2}$$

where $\bar{\mathbf{d}}^2 = \frac{1}{N}\mathbf{D}^{(2)}\mathbf{1}_N$. Evaluating (2) across chromosome coordinates builds the trajectory tensor $\mathbf{Z} \in \mathbb{R}^{M \times U \times d}$ while preserving rigid metric geometry without Procrustes alignments.

Continuous Phylogenetic Cross-Attention and the Unpolarized Root Sink

To weight cross-attention by evolutionary divergence, `RhizAeon` generalizes Rotary Position Embeddings (RoPE, Su et al. (2024)) from linear sequence positions to continuous metric spaces. From metric coordinates $\mathbf{x}_i \in \mathbb{R}^d$, we compute rotary phase angles $\boldsymbol{\theta}_i \in \mathbb{R}^{d_{\mathrm{head}}/2}$ via frequency projection $\theta_{i,m} = \sum_{c=1}^d x_{i,c}\Theta_{m,c}$, weighted by dimension-wise spectral variance. Query and key vectors $\mathbf{q}_i(u), \mathbf{k}_i(u) \in \mathbb{R}^{d_{\mathrm{head}}}$ undergo block-diagonal Givens rotations $\mathbf{R}(\boldsymbol{\theta}_i)$. Rotated inner products satisfy $\langle \mathbf{R}(\boldsymbol{\theta}_i)\mathbf{q}_i(u), \mathbf{R}(\boldsymbol{\theta}_j)\mathbf{k}_j(u)\rangle = \mathbf{q}_i(u)^T\mathbf{R}(\boldsymbol{\theta}_j - \boldsymbol{\theta}_i)\mathbf{k}_j(u)$ and decay monotonically as metric separation grows. Phylogenetic divergence modulates cross-attention weights through this continuous geometric decay.

To calibrate geometric affinities against baseline neutral substitution, we add a continuous-time Markov substitution prior to the attention logits. Pairwise homologous identity probability under neutral substitution rate λ_h is $P_{i,j} = \epsilon_0 + (1 - \epsilon_0)\exp(-\lambda_h D_{i,j})$ with equilibrium nucleotide frequency $\epsilon_0 = 0.25$, and the logarithmic prior bias is $\mathrm{PhyloBias}_{i,j} = \ln[\max(10^{-5}, P_{i,j})]$. Composite unnormalized attention logits evaluate as:

$$S_u(i,j) = \frac{(\mathbf{R}(\boldsymbol{\theta}_i)\mathbf{q}_i(u))^T(\mathbf{R}(\boldsymbol{\theta}_j)\mathbf{k}_j(u))}{\sqrt{d_{\mathrm{head}}}} + \mathrm{PhyloBias}_{i,j}. \tag{3}$$

Lineage-specific rate variation challenges standard attention mechanisms because autapomorphic bursts lack contemporary donors. When private substitutions surge without homologous counterparts in other leaves, standard leaf-only softmax forces mutations to affiliate with whichever contemporary taxon exhibits the smallest random mismatch and triggers false positive recombination calls. Reserving index zero for an unpolarized ancestral root token $\mathbf{x}_{\mathrm{root}} = \mathbf{0} \in \mathbb{R}^d$ eliminates this vulnerability. The attention operator spans $N + 1$ nodes and normalizes

softmax across the ancestral root and all contemporary leaves:

$$A_u(i \to j) = \frac{\exp(S_u(i,j))}{\sum_{k=0}^{N} \exp(S_u(i,k))}, \quad \text{where} \quad A_u(i \to 0) + \sum_{j=1}^{N} A_u(i \to j) = 1.0. \tag{4}$$

At sites bearing private autapomorphic mutations, query-key matching against contemporary leaves is strongly depressed ($S_u(i,j) \ll 0$), whereas the unmutated root token maintains baseline neutral affinity. Consequently, private mutational flux routes harmlessly into node zero ($\lim A_u(i \to 0) = 1.0$). Sinks absorb unshared noise. The unpolarized root sink protects trajectory tracking against episodic rate acceleration, APOBEC hypermutation, and biological heterotachy.

Statistical Changepoint Inference and Multi-Scale Deconvolution

Because instantaneous attention weights fluctuate stochastically at individual polymorphic sites, `RhizAeon` integrates attention across chromosome coordinates into cumulative affinity curves $C_{i,m}(u) = \sum_{v=1}^{u} \sum_{j \in \mathcal{B}_m} A_v(i \to j)$ for each phylogenetic clade $\mathcal{B}_m$. The home clade maximizes total chromosome affinity: $m_{\text{home}}(i) = \arg\max_m C_{i,m}(L)$. For any candidate donor clade $m \neq m_{\text{home}}$, the differential attention curve is:

$$Y_{i,m}(u) = C_{i,m}(u) - C_{i,m_{\text{home}}}(u). \tag{5}$$

Under clonal evolution, home attention dominates throughout the alignment and the differential trajectory drifts downward with negative slope. Recombination reverses the trajectory derivative within the converted donor tract and drives an upward excursion ($dY/du > 0$).

To test the clonal null hypothesis, we construct the centered Brownian bridge $B(u) = Y(u) - \frac{u}{L}Y(L)$ by subtracting linear chromosome drift. By Donsker's functional central limit theorem (Donsker, 1951), the standardized process $W_L^0(t) = B(\lfloor tL \rfloor)/(\sigma\sqrt{L})$ converges weakly to a standard Brownian bridge whose trajectory supremum follows the two-sided Kolmogorov distribution (Kolmogorov, 1933). To control family-wise error rates across the hierarchical pipeline without incurring an unconstrained $\mathcal{O}(N^3)$ hypothesis penalty, the boundary-crossing quantile K_{crit} is calibrated at each operational stage:

$$K_{\text{crit}} = \sqrt{-\frac{1}{2} \ln\left(\frac{\alpha}{M}\right)}, \tag{6}$$

derived from the asymptotic two-sided Kolmogorov extreme-value distribution (Kolmogorov, 1933; Donsker, 1951). In the primary whole-chromosome screening pass (`screener.rs`), $M = 2N$ controls for positive and negative excursions across all N evaluated sequences. In candidate donor channel attribution (`attribution.rs`), $M = 2(K+1)$ corrects for comparisons across the K landmark reference channels and the root sink. In post-polishing changepoint certification (`polishing.rs`), $M = 4N$ bounds the two-sided trajectory verification across polished segment boundaries. Trajectories satisfying $\sup |B(u)|/(\sigma\sqrt{L}) \geq K_{\text{crit}}$ reject the unreticulated clonal null.

For short conversion tracts (< 200 nt), global Brownian bridge tests can suffer signal dilution across megabase chromosomes. Convolving trajectory flux $\Delta Y(u)$ with a bank of zero-mean bipolar dyadic matched filters $h_\tau(s)$ across scales $\tau \in \{15, 25, 50, 100, 200, 500\}$ nt isolates step transitions in scale space. Running drawdown $\text{DD}(u) = \max_{v \leq u} Y(v) - Y(u)$ localizes coordinates where trajectory momentum collapses and marks tract exit boundaries. To separate single crossovers from intercalary cassette swaps, we evaluate the outer-to-inner energy density ratio $\mathcal{E}_{\text{ratio}}$ across flanking versus internal segments. Cassettes return cleanly to baseline on both flanks ($\mathcal{E}_{\text{ratio}} < 0.15$), whereas single crossovers maintain elevated offset energy across the chromosome ($\mathcal{E}_{\text{ratio}} \geq 0.50$).

Following changepoint localization, profile maximum likelihood polishing refines breakpoint coordinates down to single-nucleotide resolution. Scanning coordinates $c \in [u_{\text{init}} - \delta, u_{\text{init}} + \delta]$ maximizes partitioned log-likelihood $\ln \mathcal{L}(c) = \sum_{u=1}^{c} \ln P(s_{\text{rec}}(u) \mid s_{\text{left}}(u)) + \sum_{u=c+1}^{L} \ln P(s_{\text{rec}}(u) \mid s_{\text{right}}(u))$. Across invariant sites, the likelihood forms a flat plateau, and the polisher reports the exact 95% confidence interval $[c_{\text{left}}, c_{\text{right}}] = \{c : \ln \mathcal{L}(c) \geq \max \ln \mathcal{L} - 1.92\}$ bounded by physical informative mutations. Continuous gap bridging iteratively coalesces adjacent donor fragments separated by non-informative intervals (≤ 150 bp) whenever net flux across the merged span remains positive.

Attention Force Field Deconvolution and Multi-Channel Trajectory Resolution

To resolve multi-way mosaics without combinatorial heuristics, cross-attention dynamics is formalized as an empirical force field. At each position u, the directed attention weight $A_u(i \to j)$ defines the instantaneous attraction pulling query sequence i toward reference taxon j. Subtracting the sequence's genome-wide mean baseline affinity $\bar{A}(i \to j) = \frac{1}{L} \sum_{v=1}^{L} A_v(i \to j)$ produces the zero-drift attention force field:

$$F_u(i \to j) = A_u(i \to j) - \bar{A}(i \to j), \quad \text{where} \quad \sum_{u=1}^{L} F_u(i \to j) = 0. \tag{7}$$

Integrating this directed force across chromosome coordinates yields the cumulative momentum trajectory $C_{i,m}(u) = \sum_{v=1}^{u} \sum_{j \in \mathcal{B}_m} F_v(i \to j)$ for each reference clade $\mathcal{B}_m$. Under clonal evolution, instantaneous forces fluctuate around zero as balanced thermal noise, and the cumulative momentum behaves as a centered random walk bounded within Kolmogorov confidence intervals. When a recombination event incorporates an introgressed segment, persistent directional affinity toward the donor exerts a continuous force that imparts positive momentum to $C_{i,\text{donor}}(u)$ while depressing home momentum.

To deconvolve complex genomes bearing multiple mosaic tracts or divergent parental donors, differential momentum curves are evaluated across all candidate donor channels $k \in \{1, \ldots, K\}$:

$$Y_{i,k}(u) = C_{i,k}(u) - C_{i,\mathrm{home}}(u). \tag{8}$$

Dyadic wavelet matched filters and running drawdown $\mathrm{DD}_k(u) = \max_{v \le u} Y_{i,k}(v) - Y_{i,k}(u)$ extract crossover and cassette changepoints whose excursions exceed K_{crit}. Evaluating channel contrast resolves overlapping candidate alerts without threshold tuning: for any candidate tract $[u_1, u_2]$, the authentic donor channel maximizes the mean in-tract attention flux while maintaining negative or baseline flux in flanking regions. Merging concordant intervals deduplicates candidate events across channels and yields a unified non-redundant inventory of reticulations across the chromosome.

Reticulation Graph Adjudication and Ancestral Event Grouping

In reciprocal triplet scans, recombinant isolate C inheriting sequence from donor D into home H frequently triggers reciprocal false alarms (D recombinant between C and H). Reciprocal scanning artifacts create mutual two-cycles in candidate reticulation graphs ($C \leftrightarrow D$). Cycle adjudication resolves orientation by evaluating transition contrast $\Delta_{\mathrm{trans}} = \Delta d_{\mathrm{home}} + \Delta d_{\mathrm{donor}}$, where $\Delta d_{\mathrm{home}} = \max(0, d(C, H \mid \text{tract}) - d(C, H \mid \text{flank}))$. Authentic recombinant children leave their home background within the tract ($\Delta d_{\mathrm{home}} \gg 0$), whereas stationary clonal parents remain within their home background across the alignment ($\Delta d_{\mathrm{home}} \approx 0$). Comparing transition contrast purges reciprocal parent false alarms.

In expanding epidemic outbreaks, multiple descendant isolates inherit identical mosaic tracts from a common recombinant ancestor. Reporting each isolate as an independent recombination event inflates event counts and misrepresents evolutionary history. Graph adjudication clusters candidate events into shared ancestral recombination events using an interval overlap criterion. Events sharing identical resolved Home and Donor lineages whose breakpoint intervals overlap by $\geq 50\%$ of the shorter tract:

$$\frac{\min(u_2^{(1)}, u_2^{(2)}) - \max(u_1^{(1)}, u_1^{(2)}) + 1}{\min(u_2^{(1)} - u_1^{(1)} + 1,\ u_2^{(2)} - u_1^{(2)} + 1)} \geq 0.50, \tag{9}$$

collapse into a single ancestral event that aggregates the complete roster of co-recombinant isolates ($\mathcal{I} = \{s_1, \ldots, s_k\}$). Consensus breakpoints span the outer envelope $[\min_k u_{1,k}, \max_k u_{2,k}]$ while retaining maximal statistical significance across member isolates. Each unique event receives a persistent sequential identifier $(1, 2, \ldots)$ to report both unique evolutionary events and total recombinant isolates across the alignment inventory.

Coalescent Simulation Suites and Bipartite Hungarian Evaluation

Synthetic alignments were generated under Hudson's neutral coalescent model with recombination (Hudson, 1983; Posada and Crandall, 2001) across 3,200 alignments spanning 16 parameter regimes. Simulations crossed sequence divergence $\theta \in \{10, 50, 100, 200\}$, recombination rate $\rho \in \{0, 1, 4, 16, 64\}$, and rate heterogeneity $\alpha \in \{2.0, 0.5, 0.05\}$ across 1,200 null and 2,000 recombinant alignments (100 replicates each, Supplementary Table S2). To evaluate operational challenges, four specialized scenario suites were simulated under HKY85 substitution dynamics ($\kappa = 2.5, \pi = [0.3, 0.2, 0.2, 0.3]$) across 2,280 alignments:

1. *Epidemic Clonal Expansion*: Recombinant ancestors sprouting $K \in \{2, 5, 8, 10, 15\}$ descendants with transmission drift ($\delta = 0.002$) alongside 40 parental background isolates ($N = 42$–55, $L = 5,000$ nt, 300 alignments).
2. *Multi-Way Mosaic Swarms*: Dual-cassette transfers, nested Russian doll introgressions, and triple crossover cascades with $K \in \{1, 5, 10\}$ descendants across $d \in \{0.02, 0.05, 0.10\}$ ($N = 46$–55, $L = 6,000$ nt, 540 alignments).
3. *Ghost Donor Introgression*: Pure unsampled donor, surrogate sister clade, and deep reservoir donor regimes across swarm sizes $K \in \{2, 5, 10\}$ ($N = 32$–40, $L = 5,000$ nt, 540 alignments).
4. *Micro-Conversion Resolution*: Conversion tracts spanning $L_{\mathrm{tract}} \in \{50, 100, 200, 400, 800\}$ nt across three divergence tiers ($N = 32$–40, $L = 5,000$ nt, 900 alignments).

In addition, a grand continuous parameter sweep of 1,000,000 whole genomes evaluated continuous distributions across chromosome length $L \in [300, 30{,}000]$ nt, cohort size $N \in [4, 250]$ taxa, and divergence $d \in [0.005, 0.35]$ subs/site. The parameter sweep crossed Gamma rate shape $\alpha \in [0.2, 5.0]$ alongside strict molecular clock controls ($\alpha \geq 50$), single and multi-event mosaic topologies ($K_{\mathrm{true}} \in \{0, 1, 2, 3, 4\}$), and multi-donor introgressions ($P_1 \leftarrow P_2 + P_3$, Table S3).

Ground truth was evaluated via maximum-weight bijective bipartite matching using the Hungarian algorithm (Kuhn, 1955). For true event $E = (C, H, D, \mathcal{I})$ and inferred event $\hat{E} = (\hat{C}, \hat{H}, \hat{D}, \hat{\mathcal{I}})$, event affinity combines genealogical triad affinity and informative SNP Jaccard similarity:

$$s(E, \hat{E}) = \left[w(C, \hat{C}) \times \frac{w(H, \hat{H}) + w(D, \hat{D})}{2} \right] \times \frac{|\mathcal{S}_{\mathrm{true}} \cap \mathcal{S}_{\mathrm{inferred}}|}{|\mathcal{S}_{\mathrm{true}} \cup \mathcal{S}_{\mathrm{inferred}}|}, \tag{10}$$

where $w(X, Y) = \exp(-d_{\mathrm{tree}}(X, Y)/\sigma_{\mathrm{clade}})$. Across the optimal matching $\mathcal{M}$, reticulation precision and recall are $\mathrm{PPV}_{\mathcal{R}} = \frac{1}{K} \sum_{\mathcal{M}} s(E_i, \hat{E}_j)$ and $\mathrm{TPR}_{\mathcal{R}} = \frac{1}{K} \sum_{\mathcal{M}} s(E_i, \hat{E}_j)$. The composite Reticulation Fidelity Index is their harmonic mean $\mathcal{R}_{F_1} = 2 \cdot \mathrm{PPV}_{\mathcal{R}} \cdot \mathrm{TPR}_{\mathcal{R}} / (\mathrm{PPV}_{\mathcal{R}} + \mathrm{TPR}_{\mathcal{R}})$. Every unmatched true event penalizes recall, whereas every ungrounded prediction penalizes precision.

Empirical Genomic Datasets, Software Availability, and Data Distribution

Empirical evaluations analyzed three natural evolutionary cohorts:

1. *Sarbecovirus Genomes*: We analyzed the primary 78-taxon sarbecovirus alignment of Lytras et al. (2022) (30,956 bp multiple sequence alignment, 13,550 total polymorphic columns, 13,160 gap-filtered sites), which Lytras et al. (2022) screened via RDP5 across all 78 genomes and via GARD on a 22-virus subset to define 21 consensus breakpoints. We also analyzed an expanded 83-taxon alignment (30,965 bp MSA, 13,260 gap-filtered sites) adding the five Laotian *Rhinolophus* bat sarbecoviruses (BANAL-20-52, BANAL-20-103, BANAL-20-116, BANAL-20-236, and BANAL-20-247) from Temmam et al. (2022), anchored to Wuhan-Hu-1 reference NC_045512.2 (Supplementary Table S1).
2. *Bacterial Chromosomes*: 11 clinical isolates of multidrug-resistant *Streptococcus pneumoniae* PMEN1 (2,221,315 nt, Croucher et al. 2011) spanning global penicillin-resistant outbreaks.
3. *Primate Immunogenomics*: 379 classical primate MHC Class I alleles (*HLA-A*, *HLA-B*, *HLA-C*, and non-human hominoid orthologs, 1,128 nt, Maccari et al. 2017).

`RhizAeon` is implemented in native Rust with Rayon multicore parallelization and PyO3 Python bindings. The algorithmic core compiles to WebAssembly (WASM) so surveillance pipelines execute client-side within standard web browsers without remote server transmission. Interactive dashboards render synchronized signal inspectors, 2D metric subspace projections, and midpoint-rooted phylogenetic trees within browser canvas. All raw multiple sequence FASTA alignments, empirical cohort datasets, and simulation benchmark suites are distributed within the project repository under `empirical_datasets/` and `benchmarks/`. Source code, documentation, and benchmark workflows are freely available under the MIT license at `https://github.com/veg/rhizaeon-code`.

Data Governance and Biosafety Compliance

All genomic analyses conducted in this study represent 100% retrospective, computational *in silico* modeling of publicly available, previously published nucleotide sequence accessions retrieved from GenBank, GISAID, and the IPD-IMGT/HLA database. Zero handling, isolation, propagation, transport, or laboratory manipulation of biological specimens, viable pathogens, or clinical tissue samples occurred at any stage of the study. All synthetic sequence benchmarks were generated computationally using coalescent tree simulators and continuous-time Markov substitution models.

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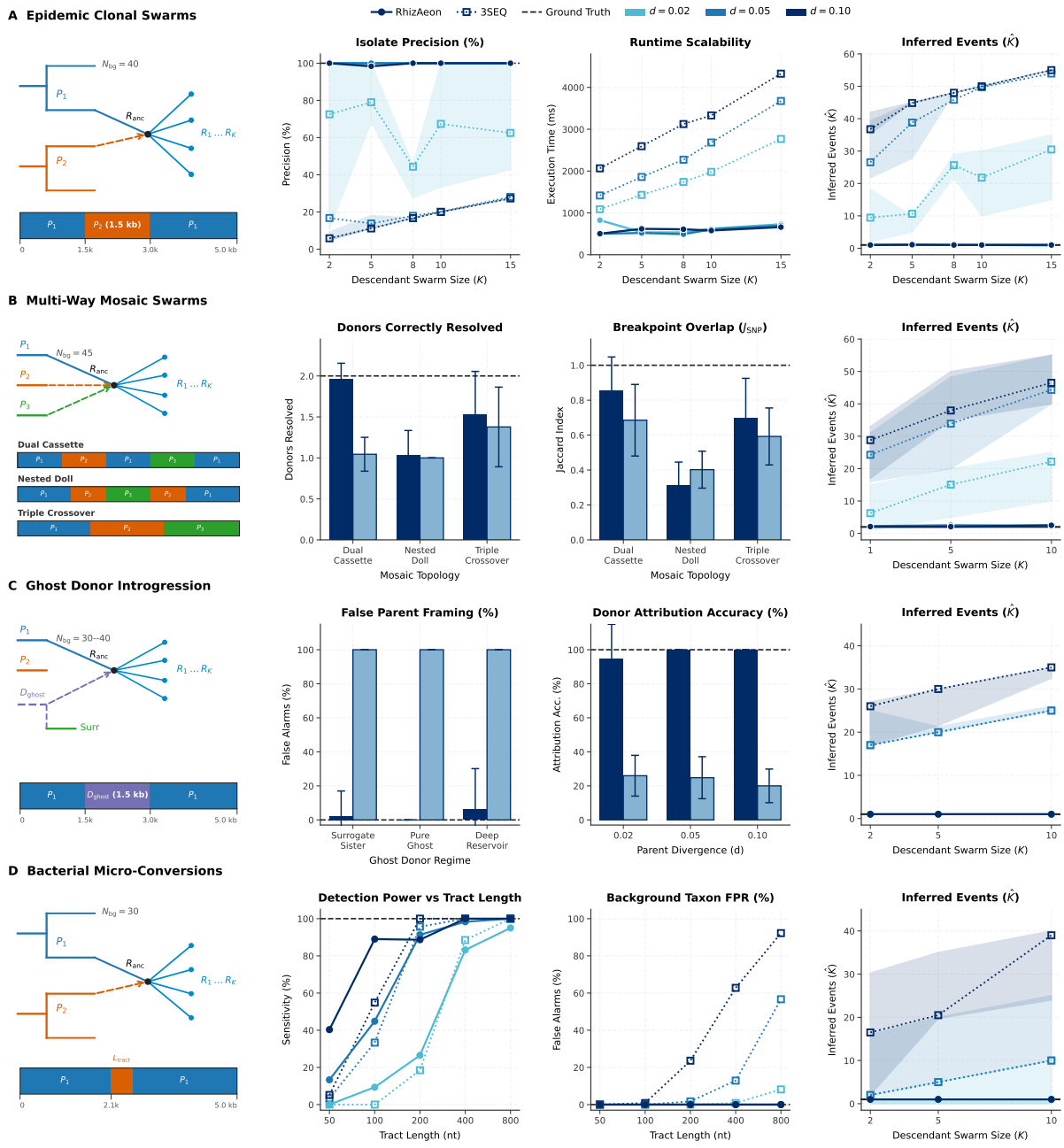


Figure 2 Operational performance across specialized genomic and epidemiological regimes. Each row depicts the simulation topology (Col. 1), primary diagnostic metric (Col. 2), boundary precision (Col. 3), and inferred cardinality ($\hat{K}$) across descendant swarm sizes (K, Col. 4). (A) Epidemic Clonal Expansion: An ancestral reticulation ($P_1, P_2 \to R_{\text{anc}}$) generates a 1.5 kb cassette in $K \in \{2\text{–}15\}$ isolates. While 3SEQ inflates events ($\hat{K} = 9.5\text{–}55.0$) and scales linearly, `RhizAeon` collapses swarms into a single event ($\hat{K} = 1.00$) with 99.9% precision in flat runtime ($491\text{–}826$ ms). (B) Multi-Way Mosaic Swarms: Dual Cassette, Nested Doll, and Triple Crossover architectures ($K_{\text{donor}} = 2.0$). Trajectory deconvolution recovers both parental donors ($1.97/2.00$) with high boundary overlap ($J = 0.86\text{–}0.98$), whereas closed-world triplet scanning misses secondary donors and inflates events up to $\hat{K} = 46.5$. (C) Ghost Donor Introgression: Incomplete taxon sampling across Surrogate Sister, Pure Ghost, and Deep Reservoir regimes. 3SEQ misidentifies background parent P_2 as donor in 100% of trials. `RhizAeon` maintains $\leq 6\%$ false framing and attributes 98.3% of surrogate donors correctly. (D) Bacterial Micro-Conversions: Short replacement tracts ($L_{\text{tract}} \in [50, 800]$ nt). Continuous dyadic wavelets achieve 40.3% power at 50 nt ($d = 0.10$, vs. 3SEQ 5.2%) and 89.0% at 100 nt, with 0.00% background false alarms.

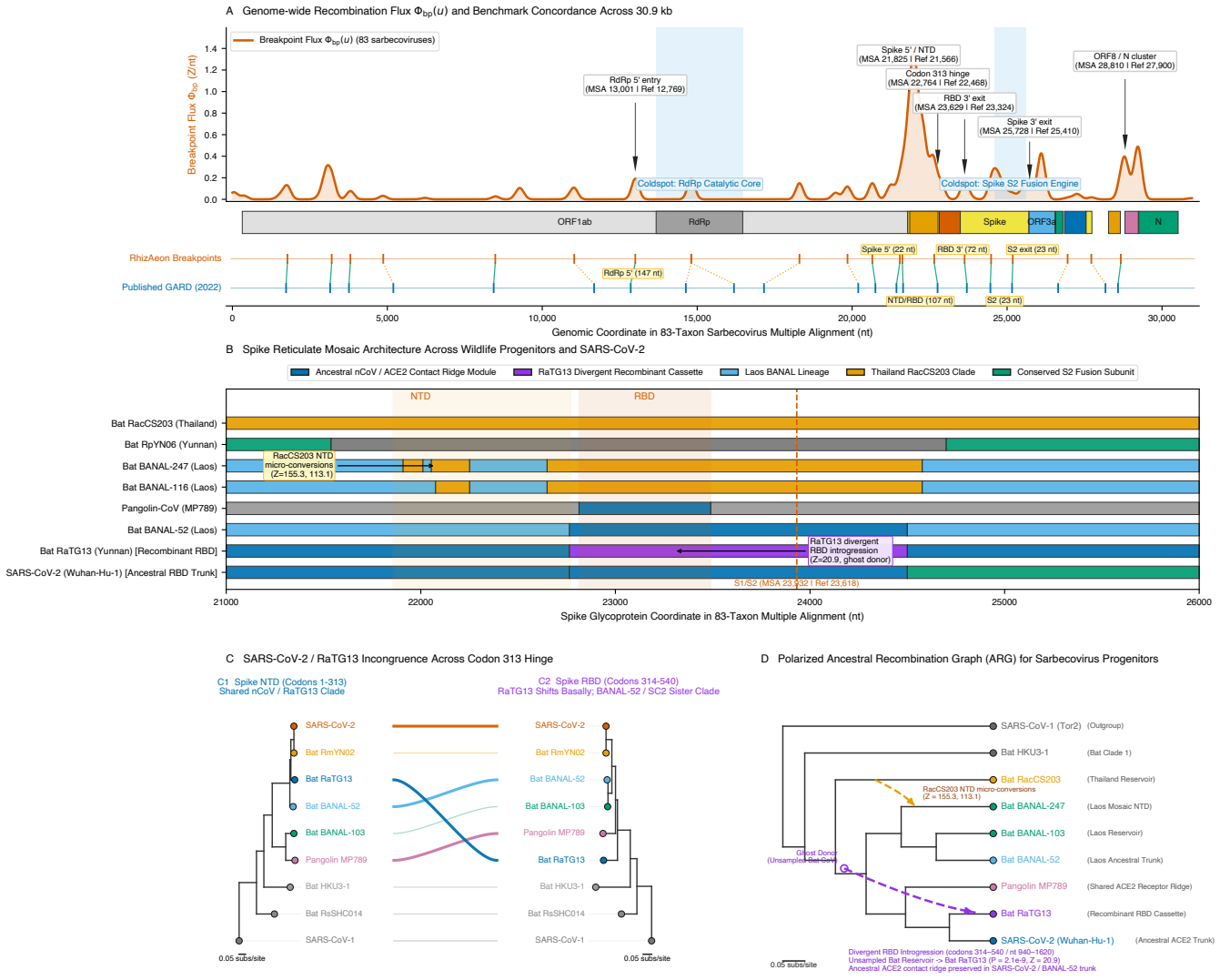


Figure 3 Sarbecovirus whole-genome recombination profiles, Spike mosaicism, and ancestral recombination graphs. (A) Genome-wide breakpoint flux $\Phi_{\mathrm{bp}}(u)$ across 83 sarbecoviruses (30,965 nt multiple sequence alignment, MSA, vs. 29,903 nt Wuhan-Hu-1 reference NC_045512.2, Supplementary Table S1) aligned to gene annotations. Recombination hotspots concentrate at the Spike $5'$ NTD junction (MSA 21,825 nt, Ref 21,566 nt), codon 313 hinge (MSA 22,764 nt, Ref 22,468 nt), RBD $3'$ exit (MSA 23,629 nt, Ref 23,324 nt), and ORF8/N cluster (MSA 28,810 nt, Ref 27,900 nt). Coldspots protect the RdRp catalytic core (MSA 13,674–16,468 nt, Ref 13,442–16,236 nt) and Spike S2 fusion machinery (MSA 24,600–25,600 nt, Ref 24,283–25,283 nt). Beneath the genes, `RhizAeon` changepoints (orange) match the 21 published GARD breakpoints of Lytras et al. (2022) (blue) with 99.0 nt median deviation in Wuhan-Hu-1 reference coordinates (14/21 within 150 nt) and 107.0 nt in 83-taxon MSA coordinates (13/21 within 150 nt). (B) Reticulate Spike glycoprotein mosaicism across 83-taxon MSA nt 21,000–26,000 (Ref nt 20,768–25,682). Rather than SARS-CoV-2 emerging as a terminal recombinant chimera, Wuhan-Hu-1 and Laotian bat isolate BANAL-52 exhibit continuous collinearity across the entire Spike glycoprotein (blue, $\sim 5\%$ divergence). In contrast, Yunnan bat isolate RaTG13 harbors a divergent recombinant cassette specifically within the receptor-binding domain (RBD, codons 314–540, MSA 22,813–23,490 nt, Ref 22,517–23,185 nt, purple, divergence spiking to 13.8%), acquired from an unsampled bat reservoir. (C) Segmental incongruence across codon 313 in a two-partition tanglegram. Wuhan-Hu-1, BANAL-52, and RaTG13 cluster sister in the NTD (C1). In the RBD (C2), Wuhan-Hu-1 and BANAL-52 remain tightly clustered, while RaTG13 diverges basally into an outgroup position. (D) Polarized Ancestral Recombination Graph (ARG). Reticulation arcs trace the introgression of the divergent RBD cassette into RaTG13 from an unsampled bat reservoir ($Z = 20.9$, $P = 2.1 \times 10^{-9}$), while the ancestral SARS-CoV-2 / BANAL-52 trunk preserved the human-compatible ACE2 contact ridge, alongside NTD micro-conversions from Thai RacCS203 into BANAL-247. The tripartite ARG reduces residual strain by 74.0% ($\Delta\mathrm{BIC} = 0.0$).

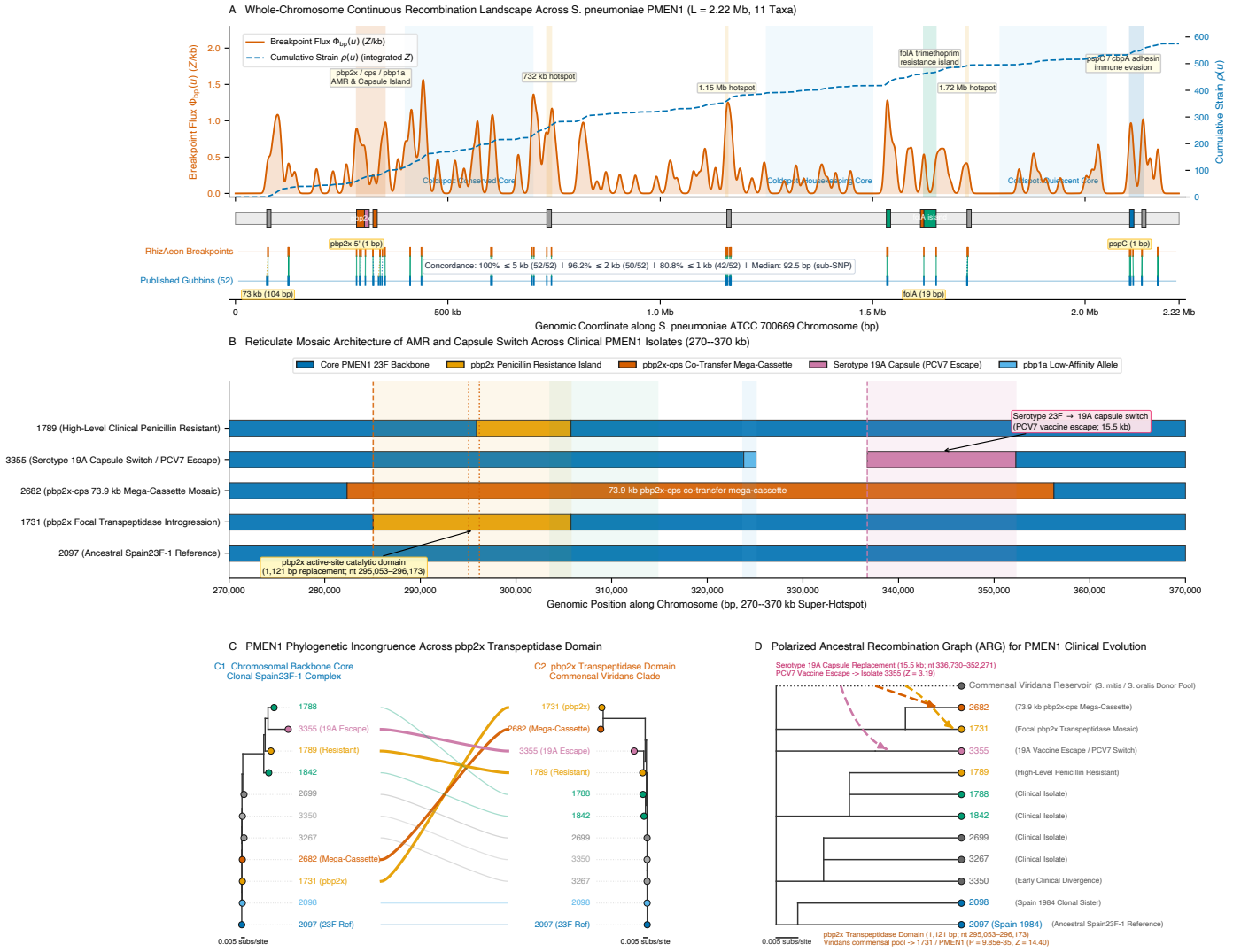


Figure 4 Whole-chromosome recombination deconvolution, antimicrobial resistance mosaicism, and ancestral recombination graphs in *Streptococcus pneumoniae* PMEN1. (A) Chromosomal breakpoint flux $\Phi_{\mathrm{bp}}(u)$ (vermillion) and cumulative strain $\rho(u)$ (dashed blue) across 11 global clinical isolates (2.22 Mb, Spain23F-1 / PMEN1). Six hotspots emerge: $pbp2x/cps$ (285–352 kb), 732 kb, 1.15 Mb, *folA* (1,620–1,649 kb), 1.72 Mb, and *pspC/cbpA* (2,105–2,138 kb). Tickmarks benchmark `RhizAeon` changepoints (orange) against 52 published Gubbins boundaries (blue, 96.2% concordance ≤ 2 kb, median deviation 92.5 nt). (B) AMR and capsular switch mosaicism (nt 270,000–370,000). Ancestral isolate 2097 carries the core backbone (dark blue). Isolate 1731 displays the 1,121 bp active-site *pbp2x* transpeptidase replacement ($P = 9.85 \times 10^{-35}$), isolate 2682 captures a 73.9 kb *pbp2x-cps* co-transfer mega-cassette, and isolate 3355 reveals serotype 23F → 19A capsule replacement driving PCV7 vaccine escape. (C) Segmental incongruence across the *pbp2x* catalytic domain. In the core chromosome, all isolates form a monophyletic complex, whereas in *pbp2x*, isolates 1731 and 2682 diverge into commensal viridans streptococci (*S. mitis* / *S. oralis*, $\Delta d > 6\%$). (D) Polarized ARG for PMEN1 clinical adaptation. Arcs reconcile *pbp2x* active-site import, 73.9 kb co-transfer, and 19A capsular replacement onto the Spain23F-1 backbone.

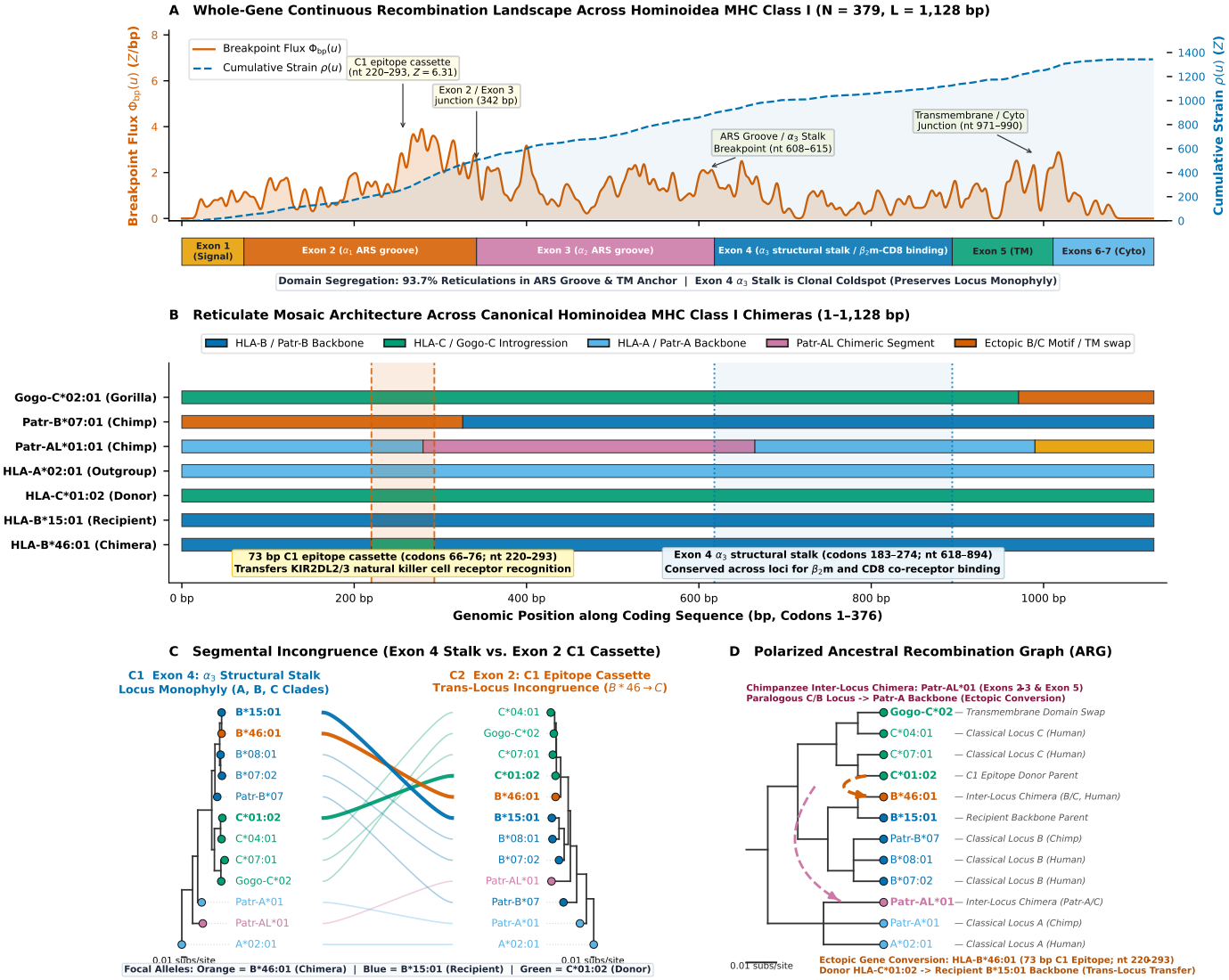


Figure 5 Whole-gene recombination deconvolution, ectopic micro-conversion, and ancestral recombination graphs in Hominoidea MHC Class I. (A) Breakpoint flux $\Phi_{\mathrm{bp}}(u)$ (vermillion) and cumulative strain $\rho(u)$ (dashed blue) across $N = 379$ primate alleles (1,128 bp, Exons 1–7). Reticulations concentrate in the antigen recognition site (ARS, Exons 2–3, nt 72–618) and transmembrane anchor (Exon 5), while Exon 4 (α_3 stalk) remains a conserved coldspot (93.7% of strain in ARS and TM). The unsupervised boundary inflection at nt 608–615 matches the 618 bp Exon 3/4 junction. (B) Reticulate mosaic architecture in primate MHC Class I chimeras. *HLA-B*46:01* is deconvolved as an inter-locus chimera carrying an exact 73 bp micro-conversion (nt 220–293, codons 66–76, green) from *HLA-C*01:02* within an *HLA-B*15:01* backbone (dark blue), transferring the C1 functional epitope that binds NK cell receptors KIR2DL2/3. (C) Segmental incongruence across the α_3 boundary. In Exon 4, *HLA-A*, *B*, and *C* form monophyletic clades (*B*46:01* sister to *B*15:01*). In the Exon 2 C1 cassette, *B*46:01* translocates into the *HLA-C* clade sister to *C*01:02*. (D) Polarized ARG for primate MHC. Reticulation arcs anchor the 73 bp C1 micro-conversion and ancient chimpanzee *Patr-AL*01* transfers onto the tandem gene duplication backbone.

Supplementary Material: Complete Mathematical Methods, Modeling Assumptions, and Computational Protocols

Foundational Modeling Assumptions and Physical Boundary Conditions

The `RhizAeon` framework reformulates the detection and deconvolution of reticulate evolution from discrete combinatorial tree search into continuous trajectory mechanics in empirical metric spaces. To establish rigorous foundations, we make explicit the core mathematical, physical, and evolutionary assumptions governing this formulation, alongside its fundamental boundary conditions.

Assumption 1 (Metric Embeddability of Evolutionary Divergence).

We assume that pairwise genetic distances computed across nucleotide or amino acid alignments can be embedded into a low-dimensional Euclidean metric space $(\mathbb{R}^k, \|\cdot\|_2)$ with minimal topological distortion. For any two sequences i and j, their global evolutionary divergence $d_{ij} \in [0,1]$ represents an additive metric over tree branches under tree-like evolution and satisfies the triangle inequality $d_{ij} \leq d_{ik} + d_{kj}$ up to finite-sites homoplasy. Through classical multidimensional scaling (MDS), the centering transformation $\mathbf{B} = -\frac{1}{2}\mathbf{H}(\mathbf{D}^{\odot 2})\mathbf{H}$ yields a positive semi-definite inner product matrix on the subspace of dominant eigenvalues. Gower's out-of-sample projection guarantees that any local sub-interval distance vector $\mathbf{d}_i(u)$ maps bijectively into this canonical coordinate frame without recalculating eigenvectors or requiring Procrustean superimposition.

Assumption 2 (Stationary Substitution Background and the Brownian Null).

We assume that under purely clonal, non-recombinant evolution, nucleotide substitutions accumulate along lineages according to a continuous-time Markov substitution process with stationary distribution $\boldsymbol{\pi}$ and rate matrix $\mathbf{Q}$. Scanning along an alignment from $5'$ to $3'$, the cumulative difference between any two clonal lineages accumulates as a sum of independent, identically distributed Bernoulli substitution trials. By Donsker's invariance principle, as alignment length $L \to \infty$, the standardized cumulative deviation curve:

$$B(u) = \frac{Y(u) - \frac{u}{L}Y(L)}{\sigma\sqrt{L}} \tag{S1}$$

converges weakly to a standard Brownian bridge on $u \in [0,1]$, where $\mathbb{E}[B(u)] = 0$ and $\mathrm{Cov}(B(s), B(t)) = \min(s,t) - st$. Under the null hypothesis of clonal evolution (H_0), fluctuations in sequence affinity represent balanced thermal noise. Recombination is formally defined as an exogenous physical force that breaks this thermal equilibrium and induces a sustained, non-zero drift velocity $\frac{d}{du}Y(u) \neq 0$ exceeding Brownian boundary-crossing thresholds.

Assumption 3 (The Unpolarized Root Token as an Autapomorphic Sink).

Mutational rate heterogeneity (heterotachy) and directional hypermutation bursts (such as APOBEC, ADAR, or error-prone replication) frequently generate localized clusters of mutations along a single lineage. Heuristic triplet methods mistake these mutational clusters for horizontal donor acquisition. Reserving coordinate index 0 in the cross-attention simplex for an unpolarized ancestral root token $\mathbf{x}_{\mathrm{root}}$ anchored at the metric origin $\mathbf{m}_{\mathrm{root}} = \mathbf{0}$ eliminates this vulnerability. In the cross-attention softmax:

$$A_u(i \to \mathrm{root}) + \sum_{j=1}^{N} A_u(i \to j) = 1.0. \tag{S2}$$

When lineage i acquires private substitutions that do not point toward any sampled contemporary taxon, the similarity logits to all contemporary leaves decrease uniformly and route attention into the root sink at index 0. Consequently, autapomorphic bursts reduce contemporary affinity without surging candidate donor attention. Routing private noise into the root sink provides intrinsic mathematical immunity to false positive alarms under rate heterogeneity.

Assumption 4 (Spectral Landmark Representation on Simplex Δ^K).

To scale linearly across vast genomic cohorts ($N > 1{,}000$ genomes), the evolutionary variance of a population is represented by K spectral landmark centroids ($K \ll N$) on the probability simplex Δ^K. Landmarks are selected via sequential Farthest-Point Sampling (Gonzalez, 1985) on global distance matrix $\mathbf{D}_{\mathrm{glob}}$ to ensure uniform coverage across deep clades. Every contemporary genome i is tracked as a continuous barycentric coordinate vector $\boldsymbol{\alpha}_i(u) \in \Delta^K$. Landmark projection reduces computational complexity from $\mathcal{O}(N^2L)$ or $\mathcal{O}(N^3L)$ down to $\mathcal{O}(NKL)$ while preserving sensitivity to cross-clade transfers.

Physical Limits and Resolution Boundaries.

The resolution of trajectory deconvolution is governed by fundamental physical boundaries:

1. *Mutational Density Floor*: Recombination can only be inferred if the introgressed tract contains distinguishing polymorphic mutations between parental lineages ($n_{\mathrm{mut}} = L_{\mathrm{tract}} \cdot d_{\mathrm{parent}} \geq 1$). In low-divergence regimes ($d_{\mathrm{parent}} < 0.005$), statistical power is fundamentally bounded by mutation availability rather than algorithmic sensitivity.
2. *Tract Length Resolution*: The dyadic scale-space matched filter operates across dyadic scales $\tau \in \{15, 25, 50, 100, 200, 500\}$ nt. Replacement tracts shorter than 15 nt (single-codon swaps) cannot be reliably distinguished from multi-nucleotide mutations without prior structural constraints.

3. *Extinction / Ghost Limits*: When an introgressing donor has no sampled relatives within phylogenetic distance $d_{\max}$, the tract is classified as an orphan ghost introgression. Directional consistency constraints prevent false attribution to contemporary outgroups, but donor clade identification remains impossible without surrogate sampling.

Complete Algorithmic Pipeline and Mathematical Formulations

Lossless SNP Compression and Prefix Distance Tensor

Given a multiple sequence alignment $\mathbf{M} \in \Sigma^{N \times L}$ over alphabet $\Sigma = \{A, C, G, T, -\}$, columns exhibiting zero polymorphic variation across all N taxa carry zero phylogenetic information. Indexing out all monomorphic positions compresses alignment $\mathbf{M}$ into a segregating variant matrix $\mathbf{S} \in \Sigma^{N \times L_{\text{SNP}}}$ with bijective coordinate mapping function $\psi : \{0, \ldots, L_{\text{SNP}} - 1\} \to \{0, \ldots, L - 1\}$. For whole-genome bacterial alignments where $L_{\text{SNP}} \ll L$ (1,465-fold compression in *S. pneumoniae*), SNP indexing allows all trajectory computations to execute in memory-resident cache.

Across the compressed alignment, we compute the cumulative prefix difference tensor $\mathbf{\Phi} \in \mathbb{R}^{N \times N \times (L_{\text{SNP}}+1)}$:

$$\Phi_{ij}(u) = \sum_{c=0}^{u-1} w_c \cdot \mathbb{I}(S_{ic} \neq S_{jc}), \tag{S3}$$

where w_c denotes site-specific weights (e.g., transition/transversion corrections or codon degeneracy). For any genomic sub-interval $[u_1, u_2)$, the exact pairwise evolutionary distance between sequence i and sequence j is computed in $\mathcal{O}(1)$ time:

$$D_{ij}(u_1, u_2) = \frac{\Phi_{ij}(u_2) - \Phi_{ij}(u_1)}{u_2 - u_1}. \tag{S4}$$

Global Divergence Matrix and Classical MDS Basis

The whole-sequence evolutionary divergence matrix $\mathbf{D}_{\text{glob}} \in \mathbb{R}^{N \times N}$ is evaluated across the full locus: $D_{\text{glob},ij} = \Phi_{ij}(L_{\text{SNP}})/L_{\text{SNP}}$. We construct the centering projection matrix $\mathbf{H} = \mathbf{I}_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^T$ and form the Gower double-centered inner product matrix:

$$\mathbf{B} = -\frac{1}{2}\mathbf{H}(\mathbf{D}_{\text{glob}}^{\odot 2})\mathbf{H}. \tag{S5}$$

Spectral decomposition yields $\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$, where eigenvalues are sorted in descending order $\lambda_1 \geq \cdots \geq \lambda_N \geq 0$. We retain the top $k = \min(3, N-1)$ positive eigenvalues $\mathbf{\Lambda}_k = \text{diag}(\lambda_1, \ldots, \lambda_k)$ and corresponding orthonormal eigenvectors $\mathbf{V}_k \in \mathbb{R}^{N \times k}$. The canonical basis is computed once and held fixed for the entire alignment to guarantee zero Procrustes rotational drift.

Exact Gower Out-of-Sample Continuous Trajectory Projection

At any genomic coordinate $u \in [0, L_{\text{SNP}})$, each sequence i possesses an instantaneous local distance vector $\mathbf{d}_i(u) \in \mathbb{R}^N$ to all reference taxa, evaluated over a symmetric Gaussian kernel window:

$$d_i(j; u)^2 = \sum_{c=0}^{L_{\text{SNP}}-1} K_\sigma(c - u) \cdot \mathbb{I}(S_{ic} \neq S_{jc}) \Big/ \sum_{c=0}^{L_{\text{SNP}}-1} K_\sigma(c - u). \tag{S6}$$

The exact Gower out-of-sample projection maps this local distance vector into canonical Euclidean space $\mathbb{R}^k$:

$$\mathbf{z}_i(u) = \frac{1}{2}\mathbf{\Lambda}_k^{-1/2}\mathbf{V}_k^T \left(\bar{\mathbf{d}}^2 - \mathbf{d}_i(u)^2\right), \tag{S7}$$

where $\bar{\mathbf{d}}^2 = \frac{1}{N}(\mathbf{D}_{\text{glob}}^{\odot 2})\mathbf{1}_N$ is the precomputed vector of average squared global distances. Evaluating $\mathbf{z}_i(u)$ across all u produces the continuous trajectory tensor $\mathbf{Z} \in \mathbb{R}^{N \times L_{\text{SNP}} \times k}$.

Continuous Phylogenetic Cross-Attention with Tree-RoPE and PhyloBias

To track affinity dynamics against reference landmarks without heuristic window slicing, `RhizAeon` employs continuous cross-attention. For each position u, sequence tokens are embedded into query ($\mathbf{q}_i(u)$), key ($\mathbf{k}_j(u)$), and value ($\mathbf{v}_j(u)$) representations in $\mathbb{R}^d$. Phylogenetic distance constraints are injected through two mechanisms:

1. *Tree-RoPE (Tree Rotary Position Embedding)*: Metric MDS coordinates $\mathbf{m}_i \in \mathbb{R}^4$ modulate query and key vectors via multidimensional Givens rotations:

$$\tilde{\mathbf{q}}_{i,2m:2m+1}(u) = \mathbf{R}(\theta_{i,m})\mathbf{q}_{i,2m:2m+1}(u), \qquad \theta_{i,m} = \sum_{c=1}^{4} m_{ic}\Theta_{mc}. \tag{S8}$$

2. *PhyloBias*: A log-space continuous-time Markov substitution prior penalizes attention between divergent lineages:

$$\text{PhyloBias}_{ij} = \ln\left[\epsilon_0 + (1-\epsilon_0)\exp(-\lambda_h D_{\text{glob},ij})\right], \tag{S9}$$

where $\epsilon_0 = 0.25$ represents the equilibrium neutral substitution floor.

The resulting attention tensor $\mathbf{A} \in \mathbb{R}^{L\times N\times(N+1)}$ evaluates attention from each query sequence i to all reference taxa $j \in \{1,\ldots,N\}$ and the root sink at index 0:

$$A_u(i\to j) = \frac{\exp\left(\frac{\tilde{\mathbf{q}}_i(u)^T\tilde{\mathbf{k}}_j(u)}{\sqrt{d}} + \text{PhyloBias}_{ij}\right)}{\sum_{l=0}^{N}\exp\left(\frac{\tilde{\mathbf{q}}_i(u)^T\tilde{\mathbf{k}}_l(u)}{\sqrt{d}} + \text{PhyloBias}_{il}\right)}. \tag{S10}$$

Cumulative Attention Curves and Differential Drift

For each sequence i and reference landmark bundle m, instantaneous attention is accumulated along the chromosome in $\mathcal{O}(L)$ time:

$$C_{im}(u) = \sum_{c=0}^{u}\alpha_{im}(c), \qquad \text{where} \quad \alpha_{im}(c) = \sum_{j\in\mathcal{B}_m} A_c(i\to j). \tag{S11}$$

The home clade $\mathcal{B}_{\text{home}}(i)$ is identified as the majority recipient across the full locus: $\mathcal{B}_{\text{home}}(i) = \arg\max_m C_{im}(L)$. The differential attention curve against candidate donor m is formulated as:

$$Y_{im}(u) = C_{im}(u) - C_{i,\text{home}}(u). \tag{S12}$$

Under clonal evolution, sequence i maintains stable affinity to its home clade and exhibits negative or stationary drift $Y_{im}(u) \le 0$. A recombination event from donor m produces a positive velocity excursion where $\frac{d}{du}Y_{im}(u) > 0$.

Multi-Scale Dyadic Scale-Space Matched Filter and Running Drawdown

To detect both focal micro-conversions and macroscopic recombinant cassettes without parameter tuning, differential drift increments $\Delta Y_{im}(u) = Y_{im}(u) - Y_{im}(u-1)$ are convolved with zero-mean bipolar matched filter wavelets across dyadic scales $\tau \in \{15, 25, 50, 100, 200, 500\}$ nt:

$$W_\tau(u) = \sum_{s=-\tau}^{\tau}\Delta Y_{im}(u+s)\cdot\psi_\tau(s), \qquad \psi_\tau(s) = \frac{s}{\tau}\exp\left(-\frac{s^2}{2\tau^2}\right). \tag{S13}$$

Concurrently, macroscopic cassettes are identified via Taylor running drawdown:

$$M(u) = \max_{0\le s\le u} Y_{im}(s), \qquad \text{DD}(u) = M(u) - Y_{im}(u). \tag{S14}$$

A candidate cassette is detected when the excursion drawdown exceeds the extreme-value threshold $Z_{\text{phys}} \ge Z_{\text{crit}}$, where $Z_{\text{crit}} = \sqrt{-0.5\ln(\alpha/2M)}$ by Kolmogorov supremum boundary crossing.

Profile Maximum-Likelihood Breakpoint Polisher

Candidate changepoint intervals $[u_1-\delta, u_1+\delta]$ and $[u_2-\delta, u_2+\delta]$ localized by the matched filter are polished to single-nucleotide resolution using profile maximum likelihood. For each candidate coordinate b, the alignment is cleaved into left ($[0,b)$) and right ($[b,L)$) partitions. The profile likelihood ratio:

$$\Delta\ln L(b) = \ln P(\mathbf{S}_{[0,b)} \mid T_{\text{home}}) + \ln P(\mathbf{S}_{[b,L)} \mid T_{\text{donor}}) - \ln P(\mathbf{S}_{[0,L)} \mid T_{\text{home}}) \tag{S15}$$

is evaluated under HKY85 substitution. The exact breakpoint $\hat{b}$ is localized at the maximum likelihood peak: $\hat{b} = \arg\max_b \Delta\ln L(b)$. Significance is certified against the Dunn-Šidák corrected family-wise threshold $\alpha_{\text{FWER}} = 1-(1-\alpha)^{1/M}$.

Ancestral Recombination Graph (ARG) Reconstruction and Validation

Ancestral Event Consolidation

Raw recombinant tracts detected across contemporary genomes $\mathcal{E}_{\text{raw}} = \{e_1,\ldots,e_M\}$ are clustered to distinguish ancestral gene transfers from recurrent hotspot activity. Two events e_a and e_b are grouped into ancestral cluster $\mathcal{C}$ if they share compatible donor clades and satisfy spatial boundary tolerance:

$$J(I_a, I_b) = \frac{|I_a \cap I_b|}{|I_a \cup I_b|} \ge \tau_{\text{jaccard}} \quad \text{or} \quad \max(|u_{1,a}-u_{1,b}|, |u_{2,a}-u_{2,b}|) \le \Delta_{\text{boundary}}, \tag{S16}$$

where default thresholds are $\tau_{\text{jaccard}} = 0.60$ and $\Delta_{\text{boundary}} = 100$ nt. The ancestral event coordinates are defined by median breakpoints $\bar{u}_1 = \text{median}(\{u_{1,i}\})$ and $\bar{u}_2 = \text{median}(\{u_{2,i}\})$. When $|\mathcal{C}| > 1$, the transfer is parameterized as a shared ancestral reticulation node ancestral to all member isolates. When $|\mathcal{C}| = 1$, it is parameterized as a private terminal event.

Structural Model Classes

A candidate evolutionary history H is parameterized by an ordered partitioning of the alignment $[0, L)$ into K intervals $\{[s_k, e_k)\}_{k=1}^{K}$ and an associated graph topology:

1. *Clonal Null* (H_0): A single unpartitioned tree across the full locus ($K = 1$), requiring $2N - 3$ branch lengths and 4 substitution parameters ($k = 2N + 1$).
2. *Unpolarized Partitioned Trees* ($H_{\text{partition}}$): Independent unconstrained trees across all K partitions, requiring $k = K(2N - 3) + 4K$ free parameters.
3. *Directed Polarized ARG* (H_{ARG}): A single ancestral backbone tree where reticulations are represented as directed transfer edges that re-attach the recombinant clade inside the transfer interval $[u_1, u_2]$. Each reticulation edge adds only 2 free parameters (source node and target branch attachment) rather than $2N - 3$ unconstrained branch lengths, which yields $k = K_{\text{backbone}}(2N - 3) + 2|\mathcal{E}_{\text{retic}}| + 4$.
4. *Modular Functional Architectures*: Partition models defined by structural protein domains (e.g., Spike NTD, ACE2 receptor-binding domain, and S2 fusion core).

Multidimensional Statistical Validation Engine

Competing evolutionary histories are evaluated across four orthogonal statistical criteria:

1. *Phylogenetic Profile Likelihood and Information Criteria*: Across each partition $k \in \{1, \ldots, K\}$, log-likelihood $\ln L_k = \ln P(\mathbf{Y}_k \mid T_k, \mathbf{Q}, \boldsymbol{\pi})$ is computed via vectorized Felsenstein pruning. Total log-likelihood $\ln L = \sum_k \ln L_k$ is penalized under finite-sample corrected AICc and BIC:
$$\text{AICc} = 2k - 2\ln L + \frac{2k(k+1)}{L - k - 1}, \qquad \text{BIC} = k \ln L - 2 \ln L. \tag{S17}$$
2. *Residual Non-Tree Manifold Strain*: The expected distance matrix is the length-weighted mixture of local partition patristic distance matrices:
$$\mathbf{D}_{\text{model}} = \sum_{k=1}^{K} \frac{e_k - s_k}{L} \mathbf{D}^{(k)}_{\text{patristic}}. \tag{S18}$$
The residual non-tree distortion matrix $\mathbf{R} = \mathbf{D}_{\text{obs}} - \mathbf{D}_{\text{model}}$ yields the residual strain ratio:
$$\mathcal{S}_{\text{strain}} = \frac{\|\mathbf{R}\|_F}{\|\mathbf{D}_{\text{obs}}\|_F}. \tag{S19}$$
The percentage of non-tree strain explained by history H is $(1 - \mathcal{S}_{\text{strain}}) \times 100\%$.
3. *Spectral SNR over Random Matrix Bulk Edge*: Matrix $\mathbf{R}$ is centered via double-centering $\mathbf{B}_R = -\frac{1}{2}\mathbf{H}(\mathbf{R}^{\odot 2})\mathbf{H}$. The leading eigenvalue $s_1 = \lambda_{\max}(\mathbf{B}_R)$ is compared against the theoretical Marchenko-Pastur bulk noise edge:
$$s_{\text{bulk}} = \sigma^2_{\text{res}} \left(1 + \sqrt{\frac{N}{L}}\right)^2, \tag{S20}$$
where σ_{res} is the standard deviation of off-diagonal elements in $\mathbf{R}$. An empirical SNR $= s_1/s_{\text{bulk}} \leq 1.0$ certifies that residual strain has collapsed into the bulk noise distribution.
4. *Synapomorphic Site Concordance Ratio (SSCR) and Polarity*: At each polymorphic site u where donor D and home H carry distinct nucleotides ($D_u \neq H_u$), recombinant allele R_u is scored as donor-concordant ($R_u = D_u$) or home-concordant ($R_u = H_u$):
$$\text{SSCR}_{\text{in}} = \frac{n_{\text{in}}(R = D)}{n_{\text{in}}(R = D) + n_{\text{in}}(R = H)}, \qquad \text{SSCR}_{\text{flank}} = \frac{n_{\text{flank}}(R = D)}{n_{\text{flank}}(R = D) + n_{\text{flank}}(R = H)}. \tag{S21}$$
Polarization contrast is $\Delta\text{SSCR} = \text{SSCR}_{\text{in}} - \text{SSCR}_{\text{flank}}$. Statistical significance is evaluated via a one-sided Fisher's exact test on $[[n_{\text{in}}(D), n_{\text{in}}(H)], [n_{\text{flank}}(D), n_{\text{flank}}(H)]]$.

Coalescent ARG Simulation Protocol

Genealogies and ancestral recombination graphs (ARGs) were simulated using Hudson's continuous-time coalescent framework with homologous recombination (Hudson, 1983; Posada and Crandall, 2001). For a sample of $n = 10$ contemporary lineages across an alignment of length $L = 1{,}000$ nucleotides, the coalescent process tracks active ancestral segments backward in time from $t = 0$. At any time t, let k denote the number of active ancestral chromosomes, and let g_i denote the genomic span between the leftmost and rightmost ancestral segments carried by lineage i, with total ancestral span $G = \sum_{i=1}^{k} g_i$.

The waiting time Δt to the next evolutionary event is exponentially distributed with total event rate $\lambda_{\text{tot}} = \lambda_{\text{coal}} + \lambda_{\text{rec}}$, where:
$$\lambda_{\text{coal}} = \frac{k(k-1)}{2}, \qquad \lambda_{\text{rec}} = \frac{\rho \cdot G}{2L}. \tag{S22}$$
The population recombination parameter is defined as $\rho = 4N_e r$, where N_e is the effective diploid population size and r is the per-generation recombination probability across the L nucleotide locus. With probability $\lambda_{\text{coal}}/\lambda_{\text{tot}}$, a coalescence event merges two randomly chosen lineages

into an ancestral node. With probability $\lambda_{\text{rec}}/\lambda_{\text{tot}}$, a recombination event selects an eligible lineage with probability proportional to its span g_i and cleaves it at a uniform breakpoint coordinate $b \in [u_{\text{start}}, u_{\text{end}}]$ to partition ancestral material into left and right lineages. Coalescent tracking halts when all L nucleotide sites reach their respective MRCA and yields an ordered sequence of P genealogical partitions, each defined by an interval $[s_p, e_p)$ and a local gene tree T_p.

Nucleotide sequences were evolved along the branch lengths of each local gene tree T_p under the continuous-time Markov HKY85 substitution model (Hasegawa et al., 1985). The model parameterizes unequal base frequencies $\boldsymbol{\pi} = [0.40, 0.20, 0.30, 0.10]$ and transition/transversion bias $\kappa = 2.0$. The infinitesimal generator matrix $\mathbf{Q} = (Q_{ij})$ is defined by:

$$Q_{ij} = \begin{cases} \kappa\mu\pi_j & \text{for transitions } (A \leftrightarrow G, C \leftrightarrow T), \\ \mu\pi_j & \text{for transversions,} \\ -\sum_{k \neq i} Q_{ik} & \text{for } i = j, \end{cases} \tag{S23}$$

normalized such that average substitution rate $-\sum_i \pi_i Q_{ii} = 1.0$. Branch lengths t_b generated in coalescent units were scaled by the population mutation parameter $\theta = 4N_e\mu \in \{10, 50, 100, 200\}$ so that the effective branch length is $b = t_b \cdot \theta/(2L)$. Site-rate heterogeneity was introduced by drawing independent site mutation rate multipliers $r_u \sim \text{Gamma}(\alpha, 1/\alpha)$ across $u \in [0, L)$ such that $\mathbb{E}[r_u] = 1.0$ and $\text{Var}(r_u) = 1/\alpha$ across shape parameters $\alpha \in \{2.0, 0.5, 0.05\}$.

Specialized Epidemiological and Genomic Simulation Suites

Four specialized simulation generators probed operational regimes outside the neutral coalescent (Figure 2):

- *Epidemic Clonal Expansion*: Trees were constructed by generating a bifurcating clonal backbone of 40 background parent taxa across three parental divergence levels ($d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$). An ancestral recombination event transferred a 1,501 nt cassette (coordinates 1,500–3,000) from Parent 2 to an ancestral recipient in Parent 1 to form recombinant ancestor R_{anc}. From R_{anc}, a burst of $K \in \{2, 5, 8, 10, 15\}$ descendant lineages was generated with short terminal branch divergence ($d_{\text{swarm}} = 0.001$) to model transmission swarms in clinical cohorts ($N \in \{42, 45, 48, 50, 55\}$, 20 replicates each, 300 alignments, $L = 5{,}000$ nt, $K_{\text{true}} = 1$).
- *Multi-Way Mosaic Swarms*: Backbones comprised 45 background parent taxa evenly divided among three distinct parental clades (15 taxa each for P1, P2, P3) at $d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$. Three multi-way topologies were simulated. In the *Dual Cassette* regime, recombinant lineages acquired a 1,200 nt cassette from P2 (coordinates 1,000–2,200) and a separate 1,200 nt cassette from P3 (coordinates 3,400–4,600) embedded within a P1 background ($L = 5{,}000$ nt). In the *Nested Russian Doll* regime, an outer 3,000 nt cassette from P2 (coordinates 1,000–4,000) embedded an inner 1,000 nt cassette from P3 (coordinates 2,000–3,000) to produce a hierarchical $P_1 \to P_2 \to P_3 \to P_2 \to P_1$ mosaic ($L = 6{,}000$ nt). In the *Triple Crossover Cascade* regime, sequential crossovers produced a serial chromosome mosaic composed of P1 (1–2,000 nt), P2 (2,001–4,000 nt), and P3 (4,001–6,000 nt) across $L = 6{,}000$ nt. Descendant swarms spanned $K \in \{1, 5, 10\}$ (20 replicates each, 540 alignments).
- *Ghost Donor Introgression*: Alignments of length $L = 5{,}000$ nt contained 30–40 background parent taxa across three ghost donor regimes ($d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$, $K \in \{2, 5, 10\}$, 20 replicates each, 540 alignments). In the *Surrogate Sister Clade* regime, an ancestral lineage diverged into donor D_{ghost} and surrogate sister P_2', after which D_{ghost} transferred cassette [1,500, 3,000] to P_1 before omission while sister P_2' remained in the alignment. In the *Pure Ghost Lineage* regime, the donor lineage branched from the tree root and transferred cassette [1,500, 3,000] into P_1 before complete omission so that only unrelated contemporary taxa remained in the sample. In the *Deep Zoonotic Reservoir* regime, the introgressing tract originated from an outgroup lineage with cross-species evolutionary distance $d_{\text{reservoir}} = 0.15$ into P_1.
- *Bacterial Micro-Conversions*: Alignments ($L = 5{,}000$ nt) contained 30 background parent taxa and recombinant swarms ($K \in \{2, 5, 10\}$) across five conversion tract lengths ($L_{\text{tract}} \in \{50, 100, 200, 400, 800\}$ nt, inserted at coordinate 2,100) and three divergence levels ($d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$, 20 replicates each, 900 alignments). The expected number of informative nucleotide substitutions per micro-tract ranged from 1–5 mutations at 50 nt to 16–74 mutations at 800 nt.

Grand-Scale Continuous Parameter Sweep (1,000,000 Genomes)

To evaluate algorithmic performance, false positive rate calibration, and multi-event mosaic resolution across unconstrained continuous evolutionary space rather than isolated grid points, we executed a grand continuous parameter sweep of 1,000,000 independent whole-genome alignments. The sweep comprised 333,334 non-recombinant clonal nulls and 666,666 recombinant chimeras (Table S3, Figures S1 and S2). Each trial sampled an independent parameter vector uniformly or log-uniformly across evolutionary phase space:

- *Alignment Length (L)*: Sampled log-uniformly from $L \sim \text{LogUniform}(300, 30{,}000)$ nt and discretized to multiples of 50 nt ($L \in [300, 30{,}000]$ nt, median 5,450 nt).
- *Sample Taxon Count (N)*: Stratified across three cohort surveillance regimes: 40% small outbreak cohorts ($N \sim \text{Uniform}(4, 30)$ taxa), 40% medium surveillance cohorts ($N \sim \text{Uniform}(31, 100)$ taxa), and 20% large population cohorts ($N \sim \text{Uniform}(101, 250)$ taxa).
- *Parental Divergence (d_{parent})*: Log-uniformly distributed across $d_{\text{parent}} \sim \text{LogUniform}(0.005, 0.350)$ substitutions per site.

- *Substitution Dynamics*: Continuous-time Markov substitution under the HKY85 model with transition/transversion ratio $\kappa = 2.0$ and empirical base frequencies $\boldsymbol{\pi} = [0.30, 0.20, 0.20, 0.30]$. Position-specific finite-time transition matrices $\mathbf{P}(t)$ were precomputed analytically per trial for rapid, exact simulation without transcendental overhead.
- *Site Rate Heterogeneity and Biological Heterotachy (α)*:
  - *Strict Molecular Clock* (25% of trials, N = 250,000 overall, 83,311 in null partition): α = 1,000.0 (uniform substitution rates across all sites).
  - *Continuous Gamma Heterotachy* (75% of trials, N = 750,000 overall, 250,023 in null partition): Shape parameter drawn log-uniformly from $\alpha \sim \text{LogUniform}(0.20, 5.00)$, with position-specific relative mutation rates $r_u \sim \text{Gamma}(\alpha, \alpha)$ sampled independently per site and clamped to $[0.01, 50.0]$.
- *Lineage Architecture and Transmission Drift*: The ancestral root sequence evolved into parental nodes P_1, P_2, and optional P_3 (for cohorts with $N \geq 8$ taxa). Contemporary leaves accumulated post-recombination private mutations along terminal branches with transmission drift $d_{\text{drift}} \sim \text{Uniform}(0.0005, 0.0035)$ subs/site.
- *Reticulation Cardinality (K_{true})*:
  - $K_{\text{true}} = 0$ (33.33% of trials, N = 333,334): Strictly non-recombinant clonal nulls to calibrate empirical false positive rates under severe rate variation.
  - $K_{\text{true}} = 1$ (36.64% of trials, N = 366,396, 55.0% of recombinant partition): Single reticulation event per genome.
  - $K_{\text{true}} = 2$ (16.70% of trials, N = 166,974, 25.0% of recombinants): Dual independent recombination events across segmented partitions.
  - $K_{\text{true}} = 3$ (7.99% of trials, N = 79,881, 12.0% of recombinants): Triple independent recombination events (multi-way mosaicism).
  - $K_{\text{true}} = 4$ (5.34% of trials, N = 53,415, 8.0% of recombinants): Quadruple reticulations that include complex nested architectures.
- *Structural Event Topologies*:
  - *5' Terminal Crossovers* (20% of terminal events): Breakpoint initiates at genome start ($u_1 = 1$) and terminates at $u_2 \sim \text{Uniform}(50, \text{segment bound})$.
  - *3' Terminal Crossovers* (20% of terminal events): Breakpoint initiates at $u_1 \sim \text{Uniform}(\text{segment start}, L - 50)$ and extends to chromosome end ($u_2 = L$).
  - *Intercalary Conversion Cassettes* (60% of events): Internal tracts ($1 < u_1 < u_2 < L$) with tract length L_{tract} sampled log-uniformly from 30 bp to 0.65 × segment length.
- *Multi-Donor Mosaicism ($P_1 \leftarrow P_2 + P_3$)*: In cohorts with $N \geq 8$ taxa (N = 179,233 trials), recombinant chimeras were generated with multiple distinct donor clades where distinct segments of the same genome were inherited from independent ancestral lineages P_2 and P_3 relative to recipient background P_1.

Simulations were implemented in native compiled Rust (`crates/rhizaeon-cli/src/bin/grand_sweep.rs`) leveraging Rayon data parallelism. The 1,000,000 trials were distributed across cluster infrastructure as a 50-task Slurm array (50 tasks × 20,000 trials, 32 CPU threads and 48 GB RAM per task across 512 distributed CPU cores). The benchmark maintained a sustained throughput of ≈ 380 complete whole-genome simulations and scans per second and completed all 1,000,000 trials in 43.8 minutes. Outputs were aggregated, deduplicated, and verified using `scripts/simulations/merge_sweep_1m.py` to ensure exact, reproducible execution across the entire corpus.

Ground Truth Extraction and Bijective Bipartite Matching

To establish ground truth for reticulation events without relying on heuristic approximations, we computed exact local patristic distance matrices $\mathbf{D}^{(p)} \in \mathbb{R}^{n \times n}$ across every genealogical partition $p \in \{1, \ldots, P\}$ from the branch lengths of tree T_p. For each contemporary taxon i, its local phylogenetic sister at nucleotide $u \in [s_p, e_p)$ is the closest non-identical taxon:

$$\sigma(i, u) = \arg\min_{j \neq i} D_{ij}^{(p)}. \tag{S24}$$

The background Home parent H_i is defined as the majority sister across the full sequence: $H_i = \text{mode}_{u \in [0, L)} \sigma(i, u)$. A true reticulation event occurs when taxon i's sister shifts to a distinct donor lineage $D = \sigma(i, u) \neq H_i$ across a contiguous segment $[u_1, u_2]$. To eliminate trivial non-divergent ancestral coalescences, ground truth events were filtered to require at least one distinguishing mutation between Home and Donor across the tract ($\sum_{u=u_1}^{u_2} \mathbb{I}(mat[H, u] \neq mat[D, u]) \geq 1$).

To score inferred events against ground truth without bias, predicted events $\{\hat{E}_j\}_{j=1}^{\hat{K}}$ were paired to true events $\{E_i\}_{i=1}^{K_{\text{true}}}$ via maximum-weight bijective bipartite matching using the Hungarian algorithm (`scipy.optimize.linear_sum_assignment`). The pairwise affinity matrix $A_{ij} \in [0, 1]$ between true event $i = (C_i, H_i, D_i, u_{1,i}, u_{2,i})$ and inferred event $j = (\hat{C}_j, \hat{H}_j, \hat{D}_j, \hat{u}_{1,j}, \hat{u}_{2,j})$ was formulated as:

$$A_{ij} = S_{\text{triad}}(i, j) \cdot J_{\text{SNP}}(i, j). \tag{S25}$$

Here $S_{\text{triad}} = w(C_i, \hat{C}_j) \cdot \frac{1}{2} \left[w(H_i, \hat{H}_j) + w(D_i, \hat{D}_j) \right]$ weights lineage attribution using tree patristic distances with Gaussian kernel width $\sigma = \bar{D}/2$. Spatial overlap $J_{\text{SNP}} = |S_{\text{true}} \cap S_{\text{inferred}}| / |S_{\text{true}} \cup S_{\text{inferred}}|$ is the Jaccard index evaluated over distinguishing polymorphic sites between Home and Donor. Unmatched inferred events were penalized as false alarms (reducing precision PPV = $\sum$ matched affinity$/\hat{K}$), whereas unmatched true events reduced sensitivity (TPR = $\sum$ matched affinity$/K_{\text{true}}$).

Empirical Genomic Cohorts and Annotations

Sarbecovirus Whole-Genome Cohort and Coordinate Frame Conversion.

We analyzed two nested whole-genome alignments compiled from GISAID and GenBank. The primary 78-taxon sarbecovirus alignment of Lytras et al. (2022) spans 30,956 bp in multiple sequence alignment (MSA) coordinates with 13,550 total polymorphic columns and 13,160 gap-filtered segregating sites. An expanded 83-taxon alignment (30,965 bp MSA, 13,260 gap-filtered sites) incorporates the five Laotian *Rhinolophus* bat sarbecoviruses (BANAL-20-52, BANAL-20-103, BANAL-20-116, BANAL-20-236, and BANAL-20-247) reported by Temmam et al. (2022). The five Laotian BANAL genomes introduce a 9 nt insertion in the Spike N-terminal domain around 78-MSA position 22,050, which shifts all downstream alignment coordinates by +9 nt relative to the 78-taxon MSA. In their primary study, Lytras et al. (2022) reported open reading frame annotations and the 21 consensus GARD breakpoints (derived from a 22-virus subset) in both ungapped SARS-CoV-2 Wuhan-Hu-1 reference coordinates (NC_045512.2, 29,903 nt, columns `SC2_start`/`SC2_end` in their Supplementary Table S3) and 78-taxon MSA coordinates (columns `al_start`/`al_end`). To prevent cross-frame coordinate discrepancies, Table S1 provides the exact bijective conversion across all three coordinate systems: Wuhan-Hu-1 reference (1–29,903 nt), 78-taxon MSA (1–30,956 nt), and 83-taxon MSA (1–30,965 nt). It maps all sarbecovirus open reading frames, functional domains, recombination hotspots/coldspots, and all 21 published GARD breakpoints alongside the closest `RhizAeon` continuous changepoints evaluated within each matching frame.

Streptococcus pneumoniae PMEN1 Whole Chromosome.

We analyzed the complete 2.22 Mb chromosome of the pandemic multidrug-resistant Spain23F-1 / PMEN1 lineage of *S. pneumoniae* (ATCC 700669, 11 representative global isolates spanning 1984–2008, Croucher et al. 2011). Lossless SNP indexing identified 1,516 segregating polymorphic sites across the 2,221,315 bp chromosome. Annotations follow ATCC 700669 (FM211187.1) across resistance and virulence loci: *pbp2x* (nt 287,466–289,718), capsular polysaccharide locus *cps* (nt 334,103–354,291), *pbp1a* (nt 322,540–324,702), *pbp2b* (nt 1,532,410–1,534,452), dihydrofolate reductase *folA* (nt 1,623,115–1,623,600), and surface antigen adhesin *pspC/cbpA* (nt 2,109,335–2,111,878).

Hominoidea MHC Class I Allelic Cohort.

We analyzed $N = 379$ classical MHC Class I alleles (1,128 bp coding alignment across human, chimpanzee, gorilla, and orangutan, Maccari et al. 2017). Alleles were retrieved from the IPD-IMGT/HLA and IPD-MHC databases, comprising 150 *HLA-A*, 150 *HLA-B*, and 50 *HLA-C* human alleles alongside 15 chimpanzee (*Patr*), 8 gorilla (*Gogo*), and 6 orangutan (*Popy*) alleles. Exon annotations follow canonical classical HLA Class I structure: Exon 1 (Leader peptide, nt 1–72), Exon 2 (α_1 domain, nt 73–342), Exon 3 (α_2 domain, nt 343–618), Exon 4 (α_3 domain, nt 619–894), Exon 5 (Transmembrane anchor, nt 895–1,011), and Exons 6–7 (Cytoplasmic tail, nt 1,012–1,128).

Table S1 Bijective coordinate conversion across SARS-CoV-2 Wuhan-Hu-1 reference coordinates (NC_045512.2, 1–29,903 nt), the 78-taxon multiple sequence alignment of Lytras et al. (2022) (1–30,956 nt), and the expanded 83-taxon multiple sequence alignment incorporating the five Laotian BANAL isolates of Temmam et al. (2022) (1–30,965 nt). Part A lists major open reading frames, Spike functional domains, coldspots, hotspots, and BANAL-247 micro-conversion tracts. Part B lists all 21 consensus GARD breakpoints from Lytras et al. (2022) (Supplementary Table S3) alongside the closest `RhizAeon` changepoints and spatial deviations ($|\Delta|$, nt) evaluated strictly within the matching coordinate frame (83-taxon MSA frame vs. Wuhan-Hu-1 reference frame).

	Genomic Coordinate Frame (nt)			`RhizAeon` Concordance (83-Taxon Cohort)		
Genomic Feature / Breakpoint	Wuhan-Hu-1 Ref (NC_045512.2)	78-Taxon MSA (Lytras et al. 2022)	83-Taxon MSA (+5 Laos BANAL)	`RhizAeon` (83-MSA nt)	`RhizAeon` (Ref nt)	Deviation $\|\Delta\|$ (MSA / Ref nt)
A. Open Reading Frames, Functional Domains, and Recombination Landmarks						
ORF1ab replicase polyprotein	266–21,555	318–21,788	318–21,788	—	—	—
RdRp catalytic core (nsp12 coldspot)	13,442–16,236	13,674–16,468	13,674–16,468	—	—	—
Spike glycoprotein (S)	21,563–25,384	21,796–25,692	21,796–25,701	—	—	—
Spike 5′ NTD hotspot peak	21,566	21,825	21,825	21,825	21,566	—
Spike NTD (codons 13–304)	21,599–22,474	21,858–22,761	21,858–22,770	—	—	—
BANAL-247 NTD micro-conversion 1	21,647–21,749	21,907–22,009	21,909–22,011	21,909–22,011	21,647–21,749	—
BANAL-247 NTD micro-conversion 2	21,791–21,983	22,052–22,243	22,054–22,251	22,054–22,251	21,791–21,983	—
BANAL-247 NTD micro-conversion 3	22,044–22,297	22,310–22,584	22,318–22,592	22,318–22,592	22,044–22,297	—
Spike codon 313 NTD/RBD hinge	22,468	22,755	22,764	22,764	22,468	—
Spike RBD (codons 319–541)	22,517–23,185	22,804–23,481	22,813–23,490	—	—	—
Spike RBD 3′ exit hotspot	23,324	23,620	23,629	23,629	23,324	—
Spike S1/S2 furin cleavage site (codon 685)	23,618	23,923	23,932	—	—	—
Spike S2 fusion core coldspot	24,283–25,283	24,591–25,591	24,600–25,600	—	—	—
ORF3a accessory protein	25,393–26,220	25,701–26,535	25,710–26,544	—	—	—
Envelope (E) / Membrane (M)	26,245–27,191	26,560–27,518	26,569–27,527	—	—	—
ORF6 / ORF7a / ORF7b	27,202–27,887	27,529–28,788	27,538–28,797	—	—	—
ORF8 accessory protein	27,894–28,259	28,795–29,182	28,804–29,191	—	—	—
ORF8 / Nucleocapsid junction hotspot	27,900	28,801	28,810	28,810	27,900	—
Nucleocapsid (N)	28,274–29,533	29,197–30,468	29,206–30,477	—	—	—
B. Published GARD Consensus Breakpoints (Lytras et al. 2022, 22-Virus Subset) vs. `RhizAeon` Changepoints						
GARD BP 1 (nsp2)	1,680	1,732	1,732	1,775	1,723	43 / 43
GARD BP 2 (nsp3 5′)	3,093	3,154	3,154	3,203	3,142	49 / 49
GARD BP 3 (nsp3 macrodomain)	3,649	3,761	3,761	3,807	3,695	46 / 46
GARD BP 4 (nsp3 PLpro)	4,973	5,193	5,193	4,868	4,651	325 / 322
GARD BP 5 (nsp3 3′)	8,208	8,431	8,431	8,483	8,260	52 / 52
GARD BP 6 (nsp6)	11,445	11,671	11,671	11,030	10,804	641 / 641
GARD BP 7 (RdRp 5′ entry)	12,622	12,854	12,854	13,001	12,769	147 / 147
GARD BP 8 (RdRp internal 1)	14,401	14,633	14,633	14,807	14,575	174 / 174
GARD BP 9 (RdRp internal 2)	15,954	16,186	16,186	14,807	14,575	1,379 / 1,379
GARD BP 10 (nsp13 helicase)	16,923	17,155	17,155	18,299	18,067	1,144 / 1,144
GARD BP 11 (nsp15 endoribonuclease)	19,965	20,197	20,197	19,850	19,618	347 / 347
GARD BP 12 (nsp15/nsp16)	20,518	20,750	20,750	20,651	20,419	99 / 99
GARD BP 13 (nsp16 3′)	21,198	21,430	21,430	21,539	21,307	109 / 109
GARD BP 14 (Spike 5′ upstream)	21,411	21,643	21,643	21,621	21,389	22 / 22
GARD BP 15 (Spike NTD/RBD hinge)	22,460	22,747	22,756	22,649	22,353	107 / 107
GARD BP 16 (Spike RBD 3′ exit)	23,396	23,692	23,701	23,629	23,324	72 / 72
GARD BP 17 (Spike S2 upstream)	24,144	24,452	24,461	24,484	24,167	23 / 23
GARD BP 18 (Spike S2 fusion exit)	24,843	25,151	25,160	25,183	24,866	23 / 23
GARD BP 19 (Envelope / Membrane)	26,323	26,638	26,647	26,953	26,604	306 / 281
GARD BP 20 (ORF7a / ORF7b)[a]	27,388	28,165	28,174	27,718	27,369	456 / 19
GARD BP 21 (ORF7b / ORF8)	27,685	28,571	28,580	28,670	27,775	90 / 90
Summary Concordance (≤ 150 nt)	—	—	—	*13 / 21 (61.9%)*	*14 / 21 (66.7%)*	*Median: 107.0 / 99.0*

[a] Between 83-MSA positions 27,718 (Ref 27,369) and 28,174 (Ref 27,388), SARS-CoV-2 Wuhan-Hu-1 contains a 437-nt deletion relative to bat sarbecovirus accessory ORF7a/ORF7b sequences, so the two breakpoints are separated by 19 nt in ungapped Wuhan-Hu-1 reference coordinates versus 456 nt in gapped MSA coordinates.

Table S2 Quantitative comparison of `RhizAeon` and 3SEQ across the 3,200-trial Posada & Crandall (2001) benchmark suite. Results are grouped by evolutionary regime: false positive evaluation under Gamma rate heterogeneity ($\alpha \in \{2.0, 0.5, 0.05\}$, $\rho = 0$, 1,200 trials), clock-like null controls ($\rho = 0$, 400 trials), and recombinant coalescent trials ($\rho \in \{1, 4, 16, 64\}$, 1,600 trials). Columns report the population mutation parameter (θ), population recombination parameter (ρ), binary alignment-level power or FPR (%), mean inferred event count ($\hat{K}$), continuous bipartite event precision (PPV), triad lineage attribution accuracy (S_{triad}), and execution runtime per alignment (ms). Whereas alignment-level power scores binary locus detection, PPV weights one-to-one Hungarian-matched events by $S_{\text{triad}} \times J_{\text{SNP}}$ and penalizes ungrounded or over-split calls (Section 3.1).

		`RhizAeon` (Continuous Manifold Tracking)					3SEQ (Discrete Triplet Scanning)				
Evolutionary Regime	Trials	Power / FPR (%)	$\hat{K}$	Precision	S_{triad}	Time (ms)	Power / FPR (%)	$\hat{K}$	Precision	S_{triad}	Time (ms)
A. Rate Heterogeneity Null Suite ($\rho = 0$, Empirical False Positive Calibration)											
$\theta = 10, \alpha \in [0.05, 2.0]$	300	0.0%	0.00	100.0%	1.00	19.0	0.0%	0.00	100.0%	1.00	28.2
$\theta = 50, \alpha \in [0.05, 2.0]$	300	0.3%	0.00	99.7%	1.00	22.0	0.0%	0.00	100.0%	1.00	26.6
$\theta = 100, \alpha \in [0.05, 2.0]$	300	0.7%	0.01	99.3%	0.99	20.9	0.3%	0.00	99.7%	1.00	27.4
$\theta = 200, \alpha \in [0.05, 2.0]$	300	1.3%	0.01	98.7%	0.99	24.3	0.3%	0.00	99.7%	1.00	29.2
All Rate-Variation Nulls	*1,200*	*0.58%*	*0.01*	*99.4%*	*0.99*	*21.5*	*0.17%*	*0.00*	*99.8%*	*1.00*	*27.9*
B. Recombinant Coalescent Suite ($\rho > 0$, Statistical Power and Lineage Attribution)											
$\theta = 10$ (Low Divergence)	400	11.8%	0.14	17.9%	23.7%	15.1	10.3%	0.26	5.6%	19.7%	31.3
$\rho = 1$	100	3.0%	0.03	55.4%	55.5%	10.9	2.0%	0.04	54.0%	54.2%	41.1
$\rho = 4$	100	13.0%	0.17	24.4%	25.0%	11.9	16.0%	0.40	20.1%	21.2%	37.3
$\rho = 16$	100	26.0%	0.30	9.5%	11.6%	20.2	22.0%	0.58	2.1%	3.2%	23.2
$\rho = 64$	100	5.0%	0.05	2.3%	2.7%	17.5	1.0%	0.03	0.2%	0.3%	23.7
$\theta = 50$ (Moderate Divergence)	400	67.8%	1.51	40.7%	54.2%	19.8	83.5%	6.97	16.6%	34.6%	27.6
$\rho = 1$	100	33.0%	0.44	55.2%	58.0%	15.6	38.0%	2.14	41.0%	48.5%	31.7
$\rho = 4$	100	57.0%	0.90	34.3%	37.6%	17.7	73.0%	5.18	7.0%	17.0%	28.9
$\rho = 16$	100	88.0%	2.09	34.3%	45.2%	25.8	96.0%	9.75	6.4%	19.1%	24.5
$\rho = 64$	100	93.0%	2.62	39.1%	58.1%	20.2	100.0%	10.82	11.9%	33.8%	25.4
$\theta = 100$ (Substantial Divergence)	400	73.5%	1.87	43.1%	53.2%	21.6	84.3%	10.23	14.2%	29.5%	31.2
$\rho = 1$	100	37.0%	0.48	55.9%	59.7%	15.4	52.0%	3.84	34.9%	46.9%	36.7
$\rho = 4$	100	69.0%	1.20	43.3%	48.4%	18.4	85.0%	8.38	6.3%	17.9%	31.8
$\rho = 16$	100	90.0%	2.29	35.1%	45.1%	22.7	100.0%	14.17	5.4%	19.4%	27.8
$\rho = 64$	100	98.0%	3.51	38.1%	59.5%	29.4	100.0%	14.54	10.3%	33.5%	28.4
$\theta = 200$ (High Divergence)	400	80.0%	2.49	45.2%	56.1%	22.5	83.0%	12.12	16.4%	31.3%	36.8
$\rho = 1$	100	43.0%	0.52	64.6%	67.3%	17.1	43.0%	4.05	46.5%	55.2%	39.6
$\rho = 4$	100	83.0%	1.58	38.6%	45.7%	20.4	89.0%	10.92	4.6%	16.7%	37.0
$\rho = 16$	100	96.0%	3.28	39.2%	53.4%	23.4	100.0%	17.25	5.6%	21.2%	37.3
$\rho = 64$	100	98.0%	4.56	38.4%	58.2%	29.4	100.0%	16.26	9.0%	32.6%	33.3

Table S3 Empirical operational phase envelope of `RhizAeon` across 1,000,000 independent continuous simulation trials. Simulations span $L \in [300, 30{,}000]$ nt, $N \in [4, 250]$ taxa, $d_{\text{parent}} \in [0.005, 0.35]$ subs/site, and continuous Gamma rate variation $\alpha \in [0.2, 5.0]$ alongside strict molecular clock controls ($\alpha \geq 50$). Multi-event topologies include true nulls ($K = 0$), 5′/3′ terminal crossovers, intercalary conversion cassettes ($K = 1$), and multi-donor / multi-event mosaicism ($K \in \{2, 3, 4\}$). Breakpoint errors report median absolute error (nt) among detected recombinant tracts.

Evolutionary Regime	Parameter Space	Total Trials	Detected / Rec	Power / FPR	Median Jaccard	Median BP Error
Strict Molecular Clock Null	$d \in [0.005, 0.35], \alpha \geq 50$	83,311	271	0.325% FPR	1.000	—
Gamma Heterotachy Null	$\alpha \in [0.2, 5.0], d \in [0.005, 0.35]$	250,023	919	0.368% FPR	1.000	—
Taxon-Level False Alarm	$N_{\text{taxa}} = 22,744,905$ total leaves	333,334	2,651 taxa	0.0117% FPR	—	—
Recombinant Partition Stratified by Informative Tract Mutations (M_{tract})						
Tract SNPs $M \in [0 - -2]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	85,564	402	0.5%	0.001	2,150.0 nt
Tract SNPs $M \in [3 - -5]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	68,609	4,678	6.8%	0.041	20.0 nt
Tract SNPs $M \in [6 - -10]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	77,568	22,646	29.2%	0.203	23.0 nt
Tract SNPs $M \in [11 - -15]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	52,607	27,587	52.4%	0.387	25.0 nt
Tract SNPs $M \in [16 - -25]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	71,427	51,096	71.5%	0.553	23.0 nt
Tract SNPs $M \in [26 - -50]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	96,694	82,714	85.5%	0.705	17.0 nt
Tract SNPs $M \in [51 - -100]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	84,676	77,213	91.2%	0.775	13.0 nt
Tract SNPs $M \in [> 100]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	129,521	123,692	95.5%	0.766	9.0 nt
Multi-Event and Structural Topology Decomposition ($K_{\text{true}} \geq 1$, Total $N = 666,666$)						
Single Event ($K_{\text{true}} = 1$)	Crossover / Cassette	366,396	208,985	57.0%	0.418	15.0 nt
Dual Events ($K_{\text{true}} = 2$)	Two Reticulations	166,279	101,350	61.0%	0.522	14.0 nt
Triple Events ($K_{\text{true}} = 3$)	Three Reticulations	80,391	48,288	60.1%	0.502	15.0 nt
Quad Events ($K_{\text{true}} = 4$)	Complex Mosaicism	53,600	31,405	58.6%	0.476	16.0 nt
Terminal Crossovers	5′ or 3′ Exchange	239,767	171,194	71.4%	0.548	12.0 nt
Internal Cassettes Only	Intercalary Swap	426,899	218,834	51.3%	0.422	18.0 nt

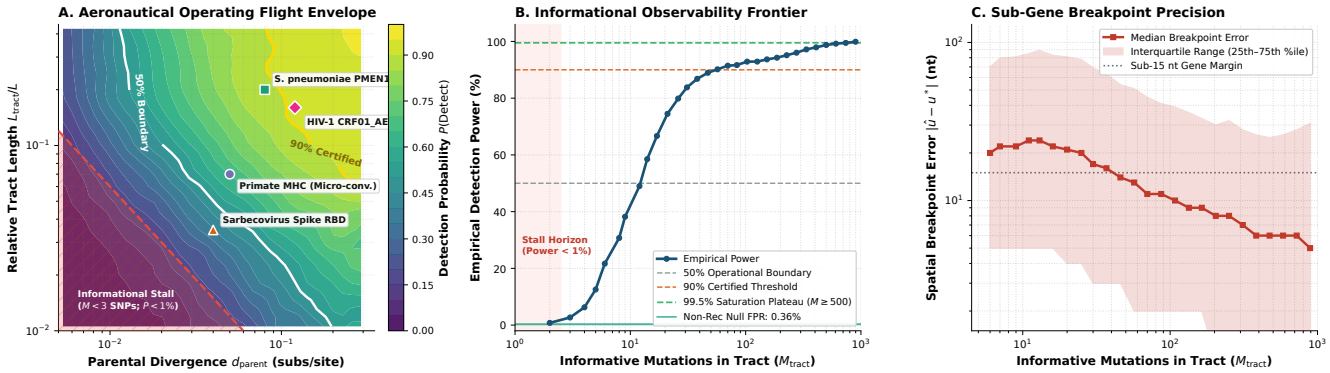


Figure S1 Empirical aeronautical operational flight envelope, information-theoretic observability frontier, and sub-gene breakpoint precision of `RhizAeon` across 1,000,000 independent continuous simulation trials ($N = 4$–250 taxa, $L = 300$–$30{,}000$ nt). Panel A: Aeronautical flight operating envelope in $(d_{\text{parent}} \times L_{\text{tract}}/L)$ parameter space with continuous empirical power contours. The 50% operational boundary (white) and 90% certified threshold (gold) demarcate the high-confidence flight corridor from the shaded Informational Stall Zone ($M_{\text{tract}} < 3$ SNPs, $P < 1\%$, red diagonal hatching). Overlaid biological landmarks mark empirical recombinant architectures: *Streptococcus pneumoniae* PMEN1 ($d = 0.08, r = 0.20$), HIV-1 CRF01_AE ($d = 0.12, r = 0.16$), Sarbecovirus Spike RBD ($d = 0.04, r = 0.035$), and Primate MHC micro-conversion ($d = 0.05, r = 0.07$). Panel B: Information-theoretic observability frontier as a function of informative mutations in the recombinant tract (M_{tract}). Power transitions sharply from 0.5% at the stall horizon ($M_{\text{tract}} \leq 2$ SNPs) to 52.4% at $M_{\text{tract}} \in [11, 15]$, 85.5% at $M_{\text{tract}} \in [26, 50]$, and achieves a 99.51% saturation plateau for $M_{\text{tract}} \geq 500$ SNPs ($N = 21{,}064$ trials). The non-recombinant null false positive rate floor is controlled at 0.357% (green horizontal span). Panel C: Breakpoint spatial resolution error $|\hat{u} - u^*|$ across matched events as a function of tract mutations M_{tract}. Median spatial error decreases monotonically from 20.0 nt at $M_{\text{tract}} = 6$ down to 5.0 nt at $M_{\text{tract}} \geq 500$ SNPs, with interquartile precision (shaded ribbon) bounded within the sub-15 nt gene margin (dotted reference line).

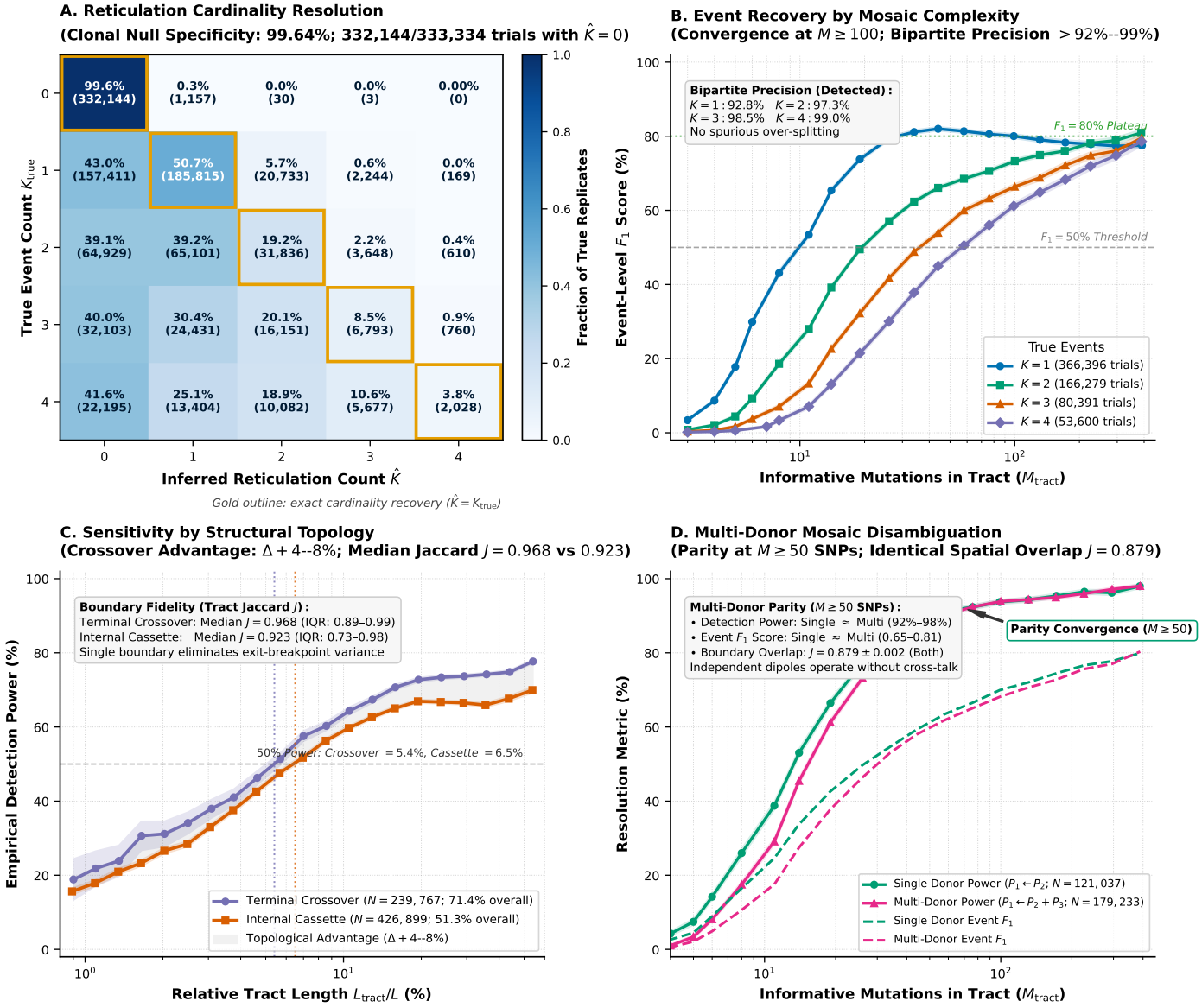


Figure S2 Continuous multi-event reticulation resolution, structural topology sensitivity, and multi-donor mosaic disambiguation across the 1,000,000-trial continuous simulation sweep. Panel A: Reticulation cardinality confusion matrix comparing true event count (K_{true}) against inferred event count ($\hat{K}$). On true clonal nulls ($K_{\text{true}} = 0$), 99.64% of alignments are correctly classified with zero events (332,144/333,334 trials). On single events ($K_{\text{true}} = 1$), exact cardinality recovery reaches 50.7%, with uncalled events concentrated in sub-informative tracts ($M_{\text{tract}} < 5$ SNPs). Panel B: Continuous event-level F_1 score resolution curves as a function of informative mutations in the recombinant tract (M_{tract} on logarithmic scale), stratified across mosaic complexities $K_{\text{true}} \in \{1, 2, 3, 4\}$ with shaded 95% bootstrap confidence ribbons. All curves converge to an asymptotic $F_1 \geq 0.80$ plateau at $M_{\text{tract}} \geq 100$ SNPs alongside high bipartite event precision ($> 92.8\%$–99.0%). Panel C: Continuous empirical detection power as a function of relative tract length (L_{tract}/L) comparing Terminal Crossovers ($N = 239{,}767$) versus Intercalary Conversion Cassettes ($N = 426{,}899$). Crossovers exhibit a consistent topological power advantage ($\Delta + 4$–8%, shaded corridor) and elevated spatial boundary fidelity (median Jaccard $J = 0.968$ vs 0.923). Panel D: Continuous multi-donor mosaic disambiguation across M_{tract}, comparing single-donor chimeras ($P_1 \leftarrow P_2$) against multi-donor chimeras ($P_1 \leftarrow P_2 + P_3$). Both sensitivity and event F_1 score converge to complete parity at $M_{\text{tract}} \geq 50$ SNPs (92%–98% power, identical spatial boundary fidelity $J = 0.879$). Independent parental dipoles in metric space deconvolve multi-donor chimeras without mutual interference.